INVENTORY MANAGEMENT SYSTEM

Warehouse, Inventory and Purchase Management

Prepared for Academic Project Documentation

Wai Lin Aung

Pearl Yadana Vocational Training Center

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CHAPTER 1

INTRODUCTION

1.1Project Description

The Inventory Management System (IMS) is a web-based application designed to support warehouse inventory and purchasing operations. The system centralizes product, category, supplier, warehouse, purchase, and stock transaction information in a structured database. It enables users to maintain inventory records, record Stock IN and Stock OUT transactions, monitor available quantities, and review purchase and transaction history.

The system follows a client-server architecture in which an Angular-based frontend communicates with Spring Boot REST APIs and a MySQL database. This architecture separates the presentation, business, and data layers and provides a foundation for maintainable and extensible inventory software.

1.2 Problem Statement

Manual inventory and purchasing processes can result in inaccurate records, duplicated data, delayed updates, and difficulty tracing stock movements. When product quantities, purchase records, and warehouse transactions are maintained manually, it becomes difficult to determine the current stock balance and identify previous transactions efficiently. These limitations can contribute to stock shortages, excess inventory, and inefficient purchasing decisions.

Therefore, a computerized inventory management solution is required to centralize records, automate stock updates, improve transaction traceability, and provide timely information for warehouse and purchasing activities.

1.3 Aim and Objectives

Aim: To design and implement a reliable web-based Inventory Management System for warehouse inventory and purchasing operations.

Objectives:

- Manage product categories and product information in a centralized system.
- Maintain supplier and warehouse records for purchasing and inventory activities.
- Create and manage purchase transactions and purchase history.
- Record Stock IN and Stock OUT transactions and update stock quantities.
- Prevent Stock OUT transactions when the available quantity is insufficient.
- Provide stock-level monitoring and transaction history for operational review.
- Maintain consistent and traceable inventory records through a centralized database.

1.4 Project Scope

The project focuses on warehouse inventory and purchasing functions. The main scope includes category management, product management, supplier management, warehouse management, purchase management, Stock IN, Stock OUT, stock monitoring, and transaction history. The system does not include laboratory management, medical records, patient management, or other unrelated business domains.

CHAPTER 2

PROJECT REQUIREMENTS

2.1 Functional Requirements

Function	Requirement
Category Management	Create, update, view, and delete product categories.
Product Management	Create and maintain product information, category assignment, pricing, stock quantity, and active status.
Supplier Management	Maintain supplier information used in purchasing and stock receiving.
Warehouse Management	Maintain warehouse information and associate inventory records with warehouse locations.
Purchase Management	Create purchase records, select suppliers and products, calculate purchase totals, confirm receipt, and view purchase history.
Stock IN	Record incoming inventory and increase the corresponding stock quantity.
Stock OUT	Record outgoing inventory after checking whether sufficient stock is available.
Stock Monitoring	Display current inventory quantities and identify low-stock conditions.
Transaction History	Review historical inventory and purchase transactions.

2.2 Non-Functional Requirements

Requirement	Description
Reliability	The system should maintain consistent inventory data during create, update, and transaction operations.
Performance	Common user actions and API requests should respond within an acceptable time.
Usability	Forms, tables, search controls, notifications, and navigation should be clear and easy to use.
Security	System access and data operations should be protected according to the application's access-control design.
Maintainability	The frontend, backend, and database layers should remain modular and understandable.
Scalability	The design should allow additional products, warehouses, suppliers, and transactions to be added without redesigning the whole system.

2.3 System Requirements

Layer	Technology / Tool
Frontend	Angular, TypeScript, PrimeNG, Tailwind CSS, RxJS
Backend	Spring Boot, Java, REST API, Hibernate/JPA
Database	MySQL Server
Development Tools	Visual Studio Code / IntelliJ IDEA, Node.js, Postman
Architecture	Client-Server REST API Architecture

CHAPTER 3

PROJECT DESIGN

3.1 System Architecture

The system is organized into three main layers: the frontend presentation layer, the backend service layer, and the database layer. The Angular frontend provides the user interface, the Spring Boot backend processes business requests through REST APIs, and MySQL stores persistent inventory and purchasing data.

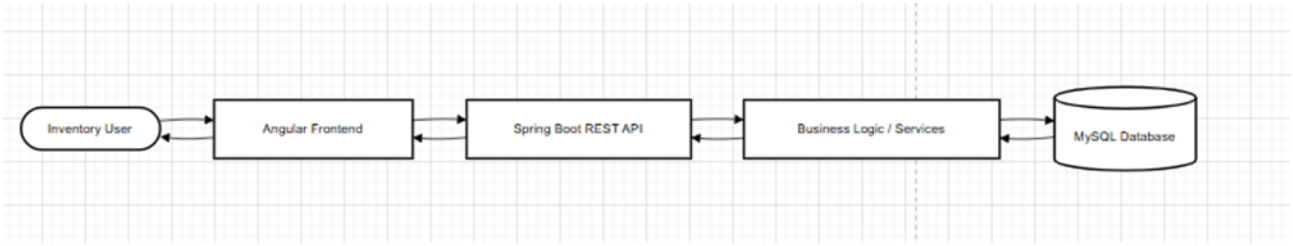


Figure 3.1: System Architecture

3.2 UML and System Diagrams

3.2.1 Use Case Diagram

The Use Case Diagram presents the major interactions between the inventory user and the system. The user can manage categories, products, suppliers, warehouses, purchases, stock transactions, stock monitoring, and transaction history.



Figure 3.2: Use Case Diagram for Inventory Management System

3.2.2 Context Diagram

The Context Diagram represents the Inventory Management System as a single system and shows its interaction with the inventory user, supplier, customer information, and the MySQL database.

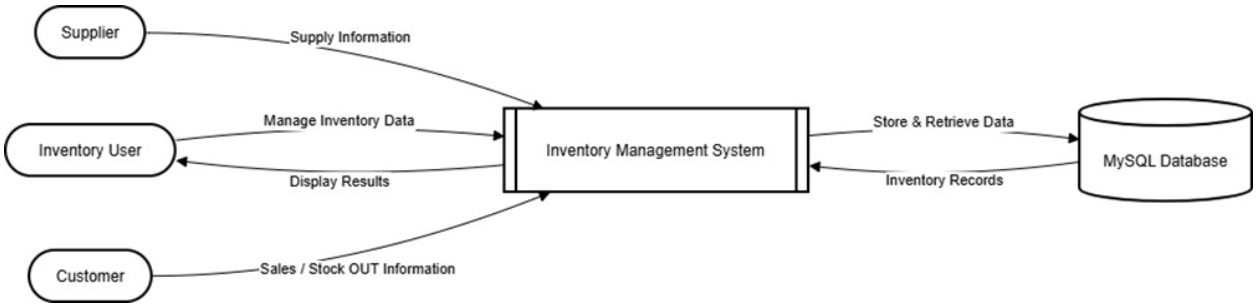


Figure 3.3: Context Diagram for Inventory Management System

3.2.3 System Overall Flowchart

The System Overall Flowchart describes the major workflow of the system. The user selects a module, performs the required operation, and the system stores valid changes in MySQL. Stock OUT includes a stock-sufficiency check before the quantity is decreased.

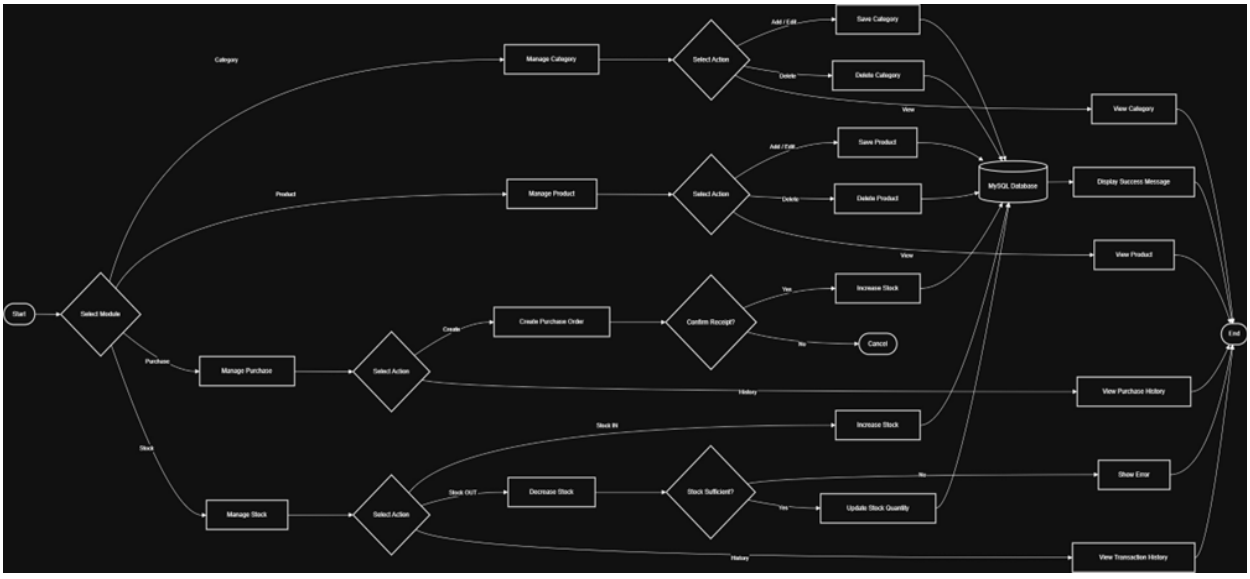


Figure 3.4: System Overall Flowchart

3.2.4 Purchase Order Flowchart

The Purchase Order Processing Flow shows the sequence from creating a purchase order to confirming receipt. When receipt is confirmed, the related inventory quantity is increased and the purchase information is retained.



Figure 3.5: Purchase Order Processing Flow

3.2.5 Stock IN/OUT Flowchart

This flow describes inventory movement. Stock IN increases inventory, while Stock OUT first verifies stock availability. An insufficient quantity results in a cancelled transaction.

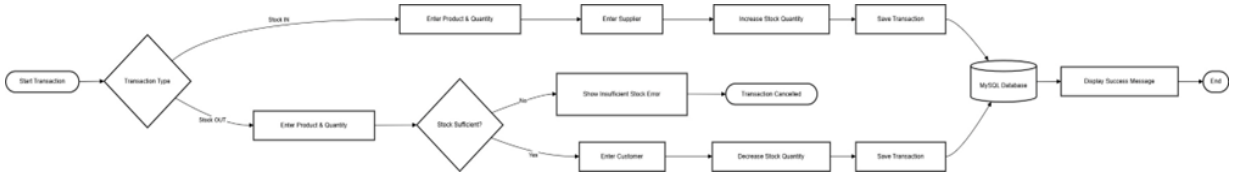


Figure 3.6: Stock IN and Stock OUT Transaction Flow

3.2.6 Purchase and Inventory Flow

This diagram explains how purchasing activities are connected to warehouse inventory. A confirmed purchase receipt results in an inventory update, while the purchase record remains available for history.

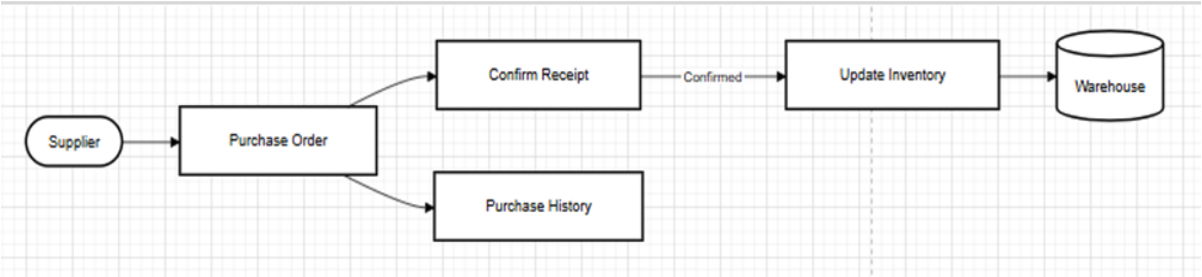


Figure 3.7: Purchase and Inventory Relationship Flow

3.3 Database Design

3.3.1 Entity Relationship Diagram

The ERD defines the relationships among the core inventory and purchasing entities. A category can contain many products, a product can have many stock transactions, suppliers can be associated with purchases, and purchase details identify the products included in each purchase.

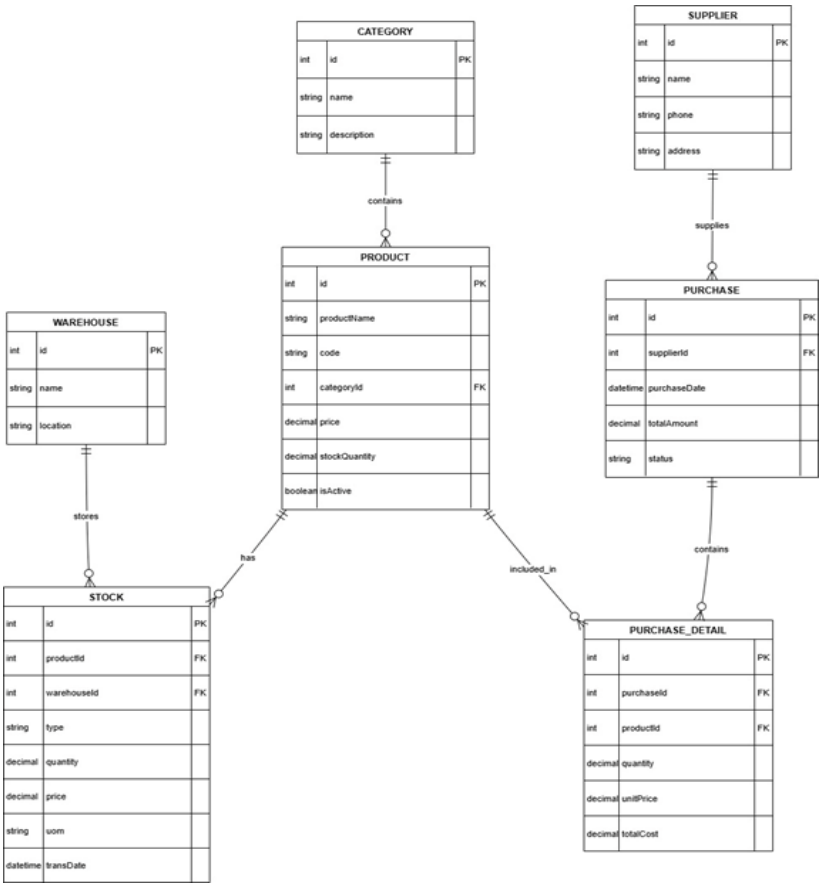


Figure 3.8: Entity Relationship Diagram

3.3.2 Data Dictionary

Table 3.1: Category Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique category identifier
name	VARCHAR(100)	NOT NULL	Category name
description	TEXT	NULL	Category description

Table 3.2: Product Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique product identifier
productName	VARCHAR(150)	NOT NULL	Product name
code	VARCHAR(50)	UNIQUE, NOT NULL	Product code
categoryId	INT	FK, NOT NULL	References CATEGORY(id)
price	DECIMAL(12,2)	NOT NULL	Unit selling or reference price

Field Name	Data Type	Constraint	Description
stockQuantity	DECIMAL(12,2)	NOT NULL	Current stock quantity
isActive	BOOLEAN	NOT NULL	Product availability status

Table 3.3: Supplier Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique supplier identifier
name	VARCHAR(150)	NOT NULL	Supplier name
phone	VARCHAR(50)	NULL	Supplier contact number
address	VARCHAR(255)	NULL	Supplier address

Table 3.4: Warehouse Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique warehouse identifier
name	VARCHAR(150)	NOT NULL	Warehouse name

Field Name	Data Type	Constraint	Description
location	VARCHAR(255)	NULL	Warehouse location

Table 3.5: Stock Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique stock transaction identifier
productId	INT	FK, NOT NULL	References PRODUCT(id)
warehouseId	INT	FK, NOT NULL	References WAREHOUSE(id)
type	VARCHAR(20)	NOT NULL	Stock IN or Stock OUT
quantity	DECIMAL(12,2)	NOT NULL	Transaction quantity
price	DECIMAL(12,2)	NOT NULL	Transaction unit price
uom	VARCHAR(30)	NOT NULL	Unit of measure
supplierName	VARCHAR(150)	NULL	Supplier for incoming stock

Field Name	Data Type	Constraint	Description
customerName	VARCHAR(150)	NULL	Customer for outgoing stock
transDate	DATETIME	NOT NULL	Transaction date and time

Table 3.6: Purchase Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique purchase identifier
supplierId	INT	FK, NOT NULL	References SUPPLIER(id)
purchaseDate	DATETIME	NOT NULL	Purchase date
totalAmount	DECIMAL(14,2)	NOT NULL	Total purchase amount
status	VARCHAR(30)	NOT NULL	Purchase status

Table 3.7: Purchase Detail Data Dictionary

Field Name	Data Type	Constraint	Description
id	INT	PK, Auto Increment	Unique purchase detail identifier
purchaseId	INT	FK, NOT NULL	References PURCHASE(id)
productId	INT	FK, NOT NULL	References PRODUCT(id)
quantity	DECIMAL(12,2)	NOT NULL	Purchased quantity
unitPrice	DECIMAL(12,2)	NOT NULL	Purchase unit price
totalCost	DECIMAL(14,2)	NOT NULL	Quantity multiplied by unit price

CHAPTER 4

IMPLEMENTATION AND USER GUIDE

This chapter describes the implementation of the Warehouse, Inventory and Purchase functions and explains how users operate each major module. The procedures are presented in a practical user-guide format so that the system workflow can be understood from data entry through database update and result display.

The implementation is organized around Category, Product, Warehouse, Supplier, Purchase, Stock IN, Stock OUT, Stock Level Monitoring, and Transaction History.

4.1 User Interface Overview

The system provides a centralized interface for managing warehouse inventory and purchasing information. Users can navigate between management modules, enter information through forms, review records in tables, and receive success or error notifications after operations.

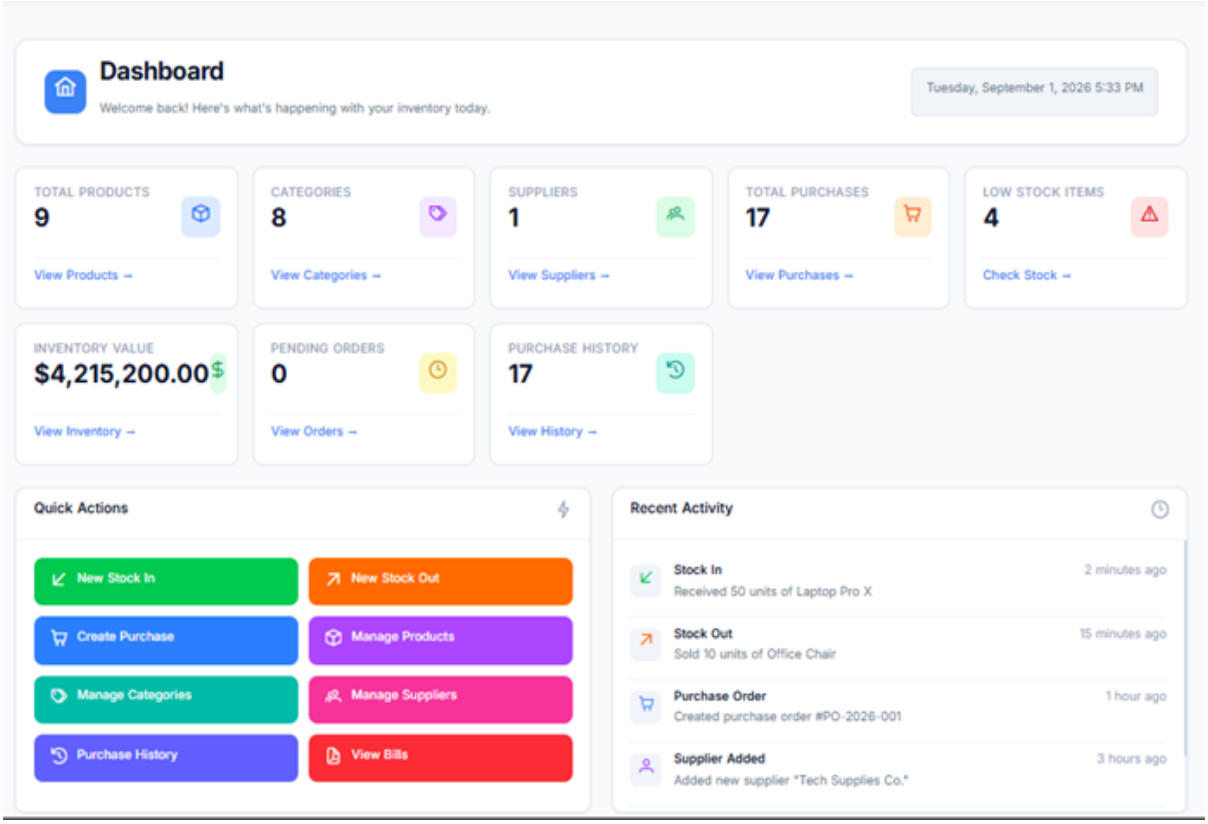


Figure 4.1: Main Inventory Management Interface

4.2 Category Management

The Category Management module is used to organize products into meaningful groups. Category records provide the structure used when creating and maintaining products.

4.2.1 View Category List

To manage categories, open the Category Management section and review the list displayed in the dynamic table. You can use the search and filtering options to quickly locate specific items, as well as check category details alongside their associated product counts.

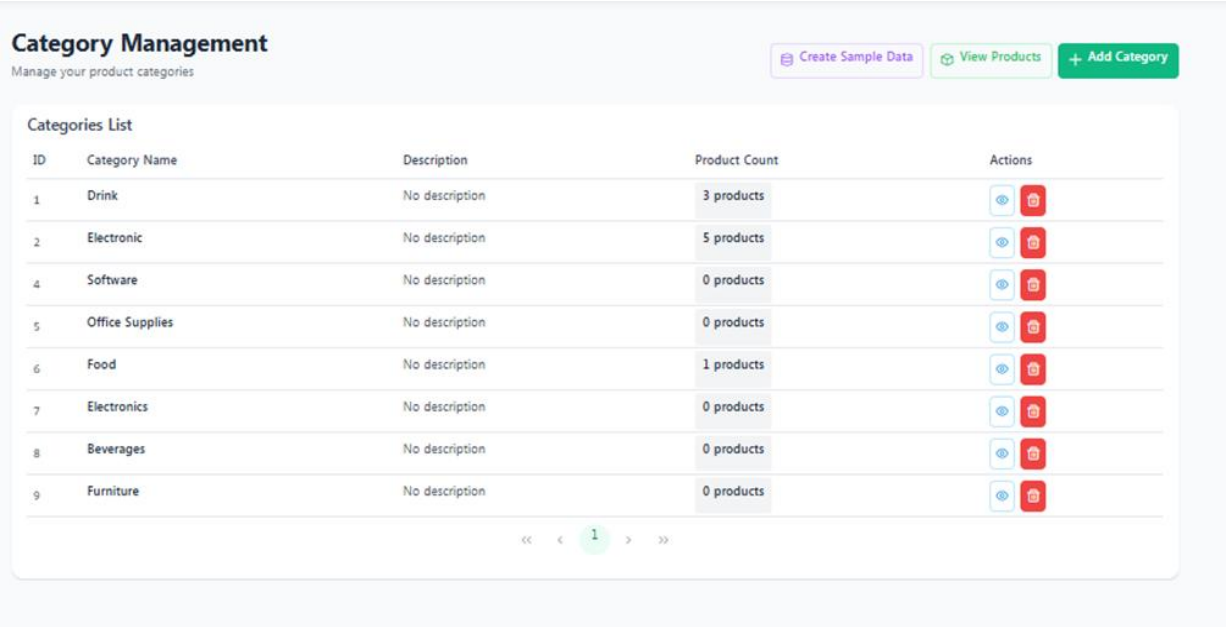
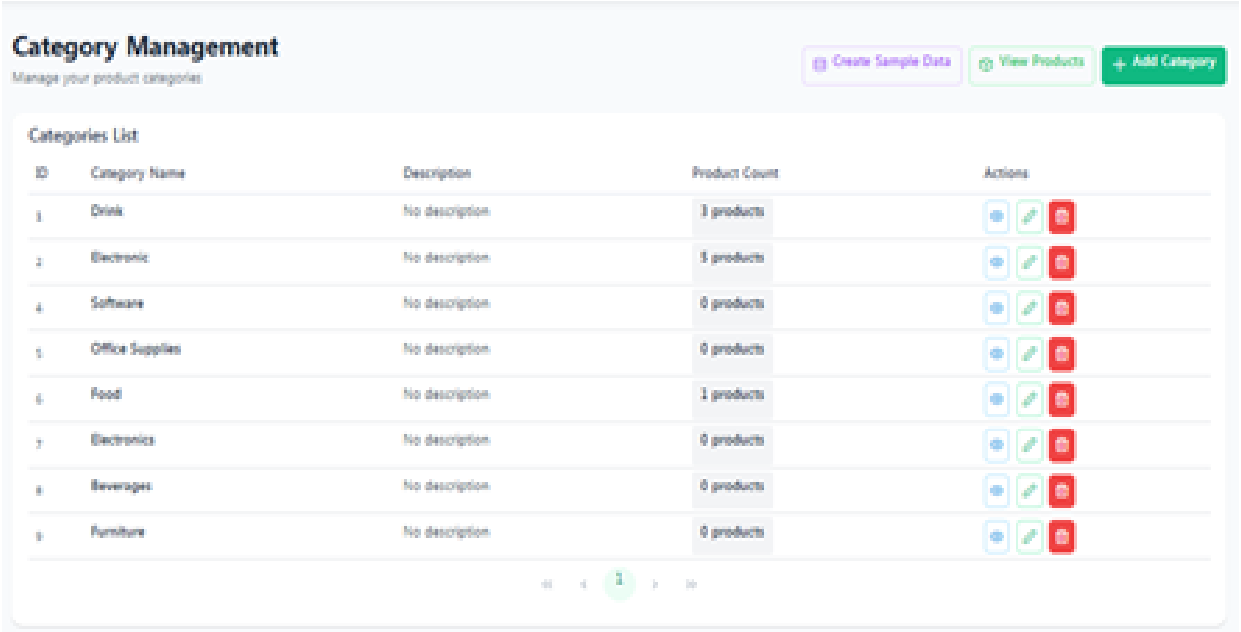
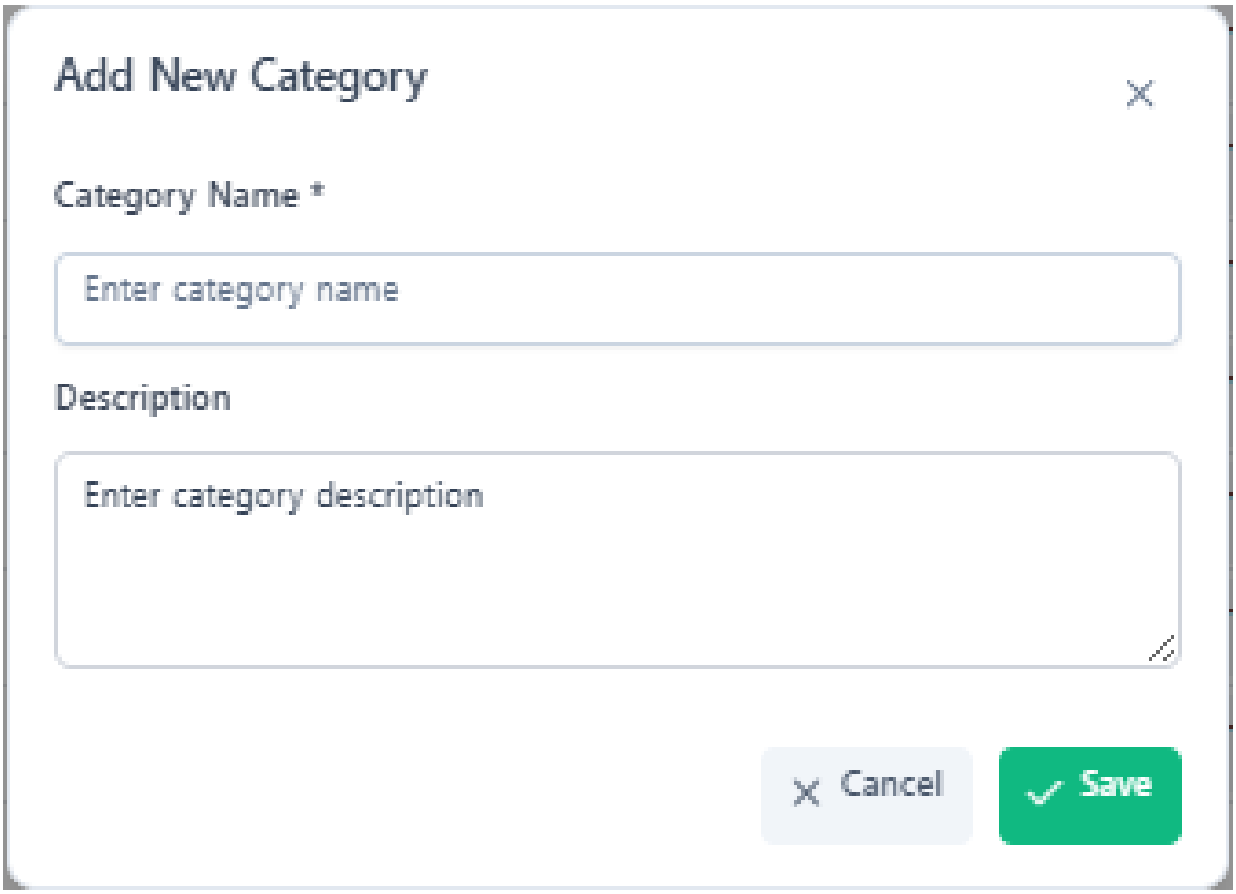


Figure 4.2: Category Management Page

4.2.2 Add Category

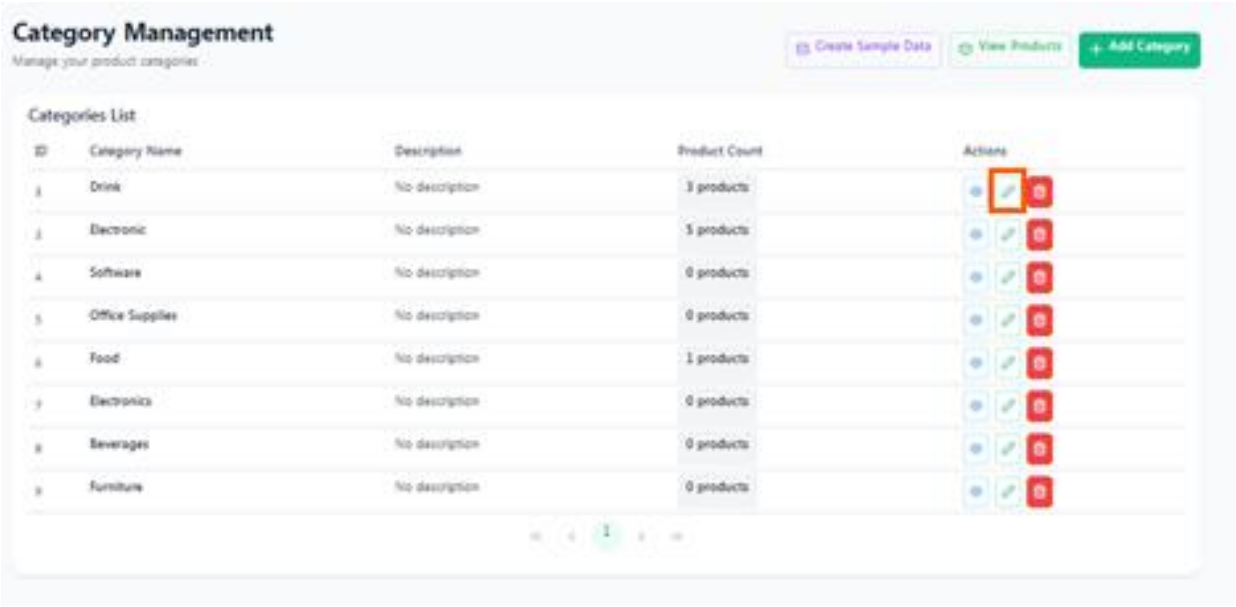
To add a new category, click Add Category and enter the Category Name along with a Description if required. Review the entered information, then click Save. Once saved, confirm the success notification and verify that the newly added category appears in the updated category list.

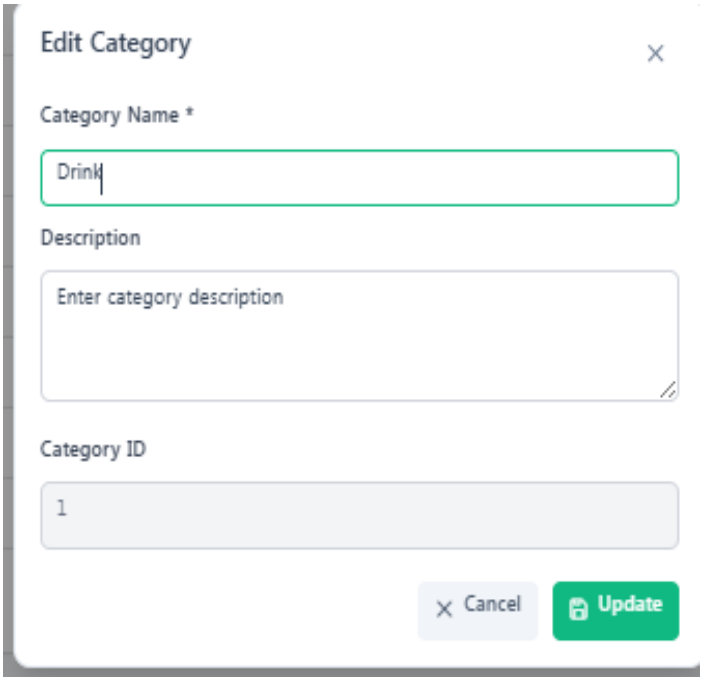

Category Management
Manage your product categories
Create Sample Data
View Products
Add Category
Categories List
ID
Category Name
Description
Product Count
Actions
1 Drink No description 3 products
2 Electronic No description 5 products
4 Software No description 0 products
5 Office Supplies No description 0 products
6 Food No description 1 products
7 Electronics No description 0 products
8 Beverages No description 0 products
9 Furniture No description 0 products


Add New Category
Category Name *
Enter category name
Description
Enter category description
Cancel
Save

4.2.3 Edit Category

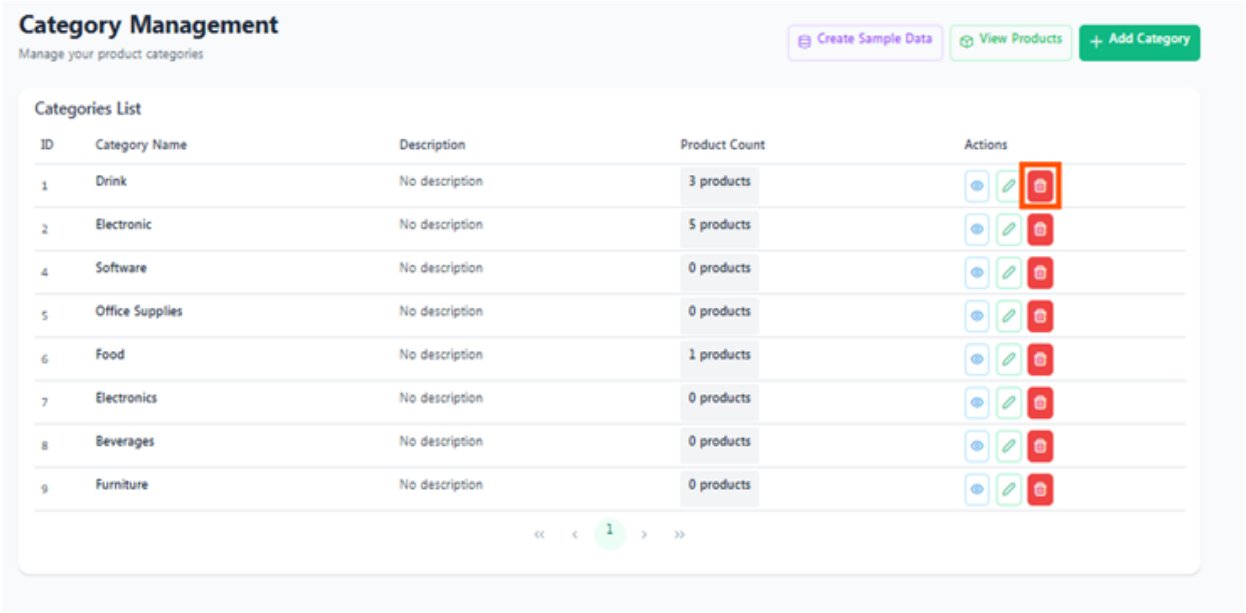
Select the required category from the list and click **Edit**. Modify the necessary details in the form, then click **Save** to apply the changes and verify that the record has been successfully updated.





4.2.4 Delete Category

Select the category you wish to remove from the list and click **Delete**. Review the confirmation prompt when displayed, then confirm the deletion to remove the record and verify that the item is no longer visible in the category list.



4.3 **Product Management**

The Product Management module maintains the products handled by the warehouse. Product records contain identifying information, category association, pricing, stock quantity, and active status.

4.3.1 View Product List

Open the Product Management section to review the full product table. You can use the search or filtering options to quickly locate specific items by code, name, or category, as well as check the displayed price, stock quantity, and active status for each product.

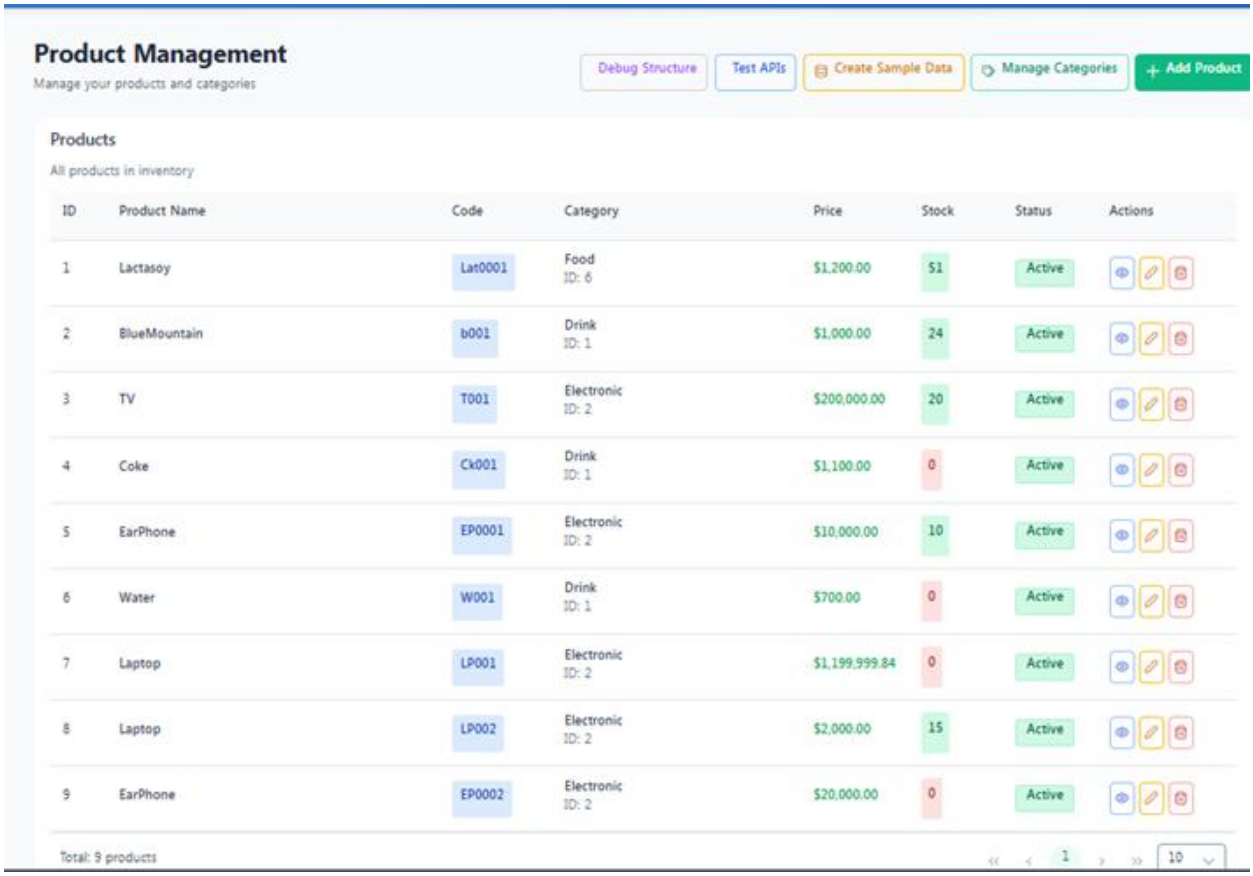
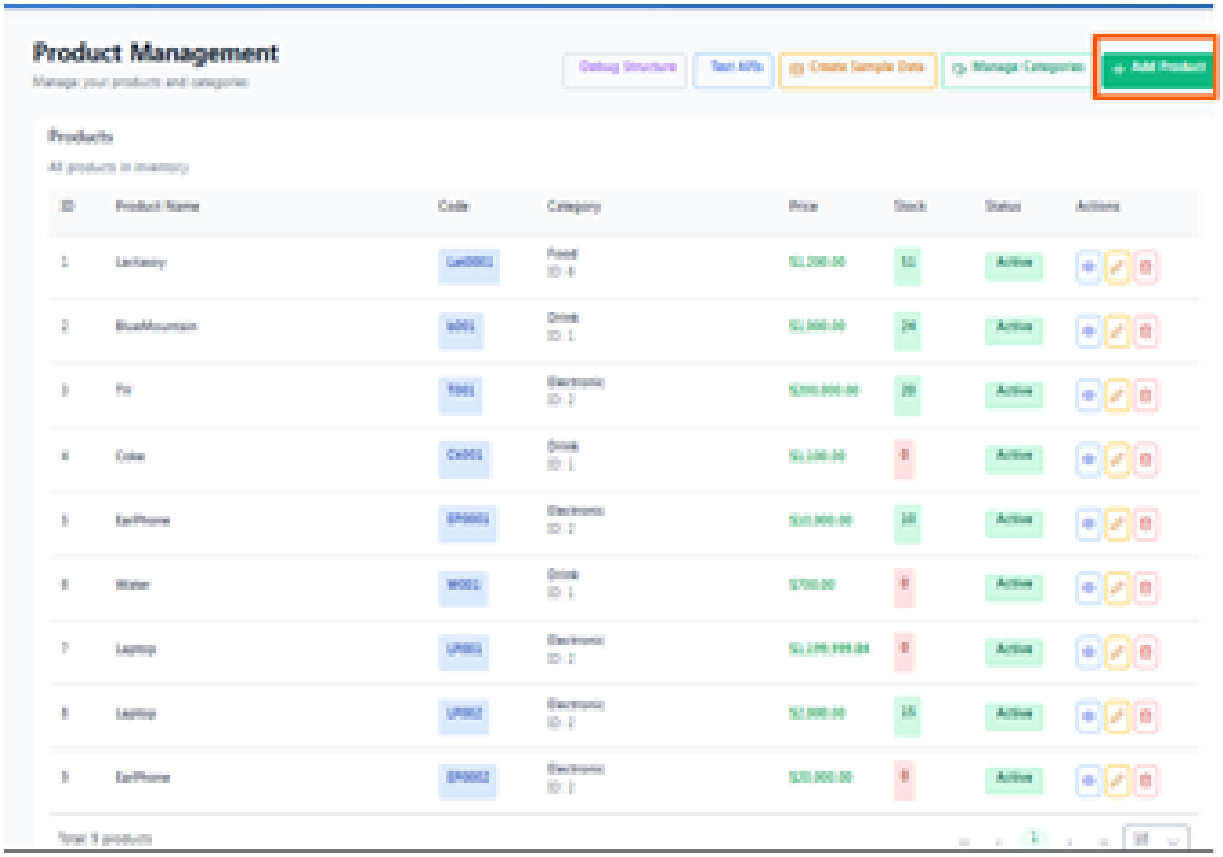
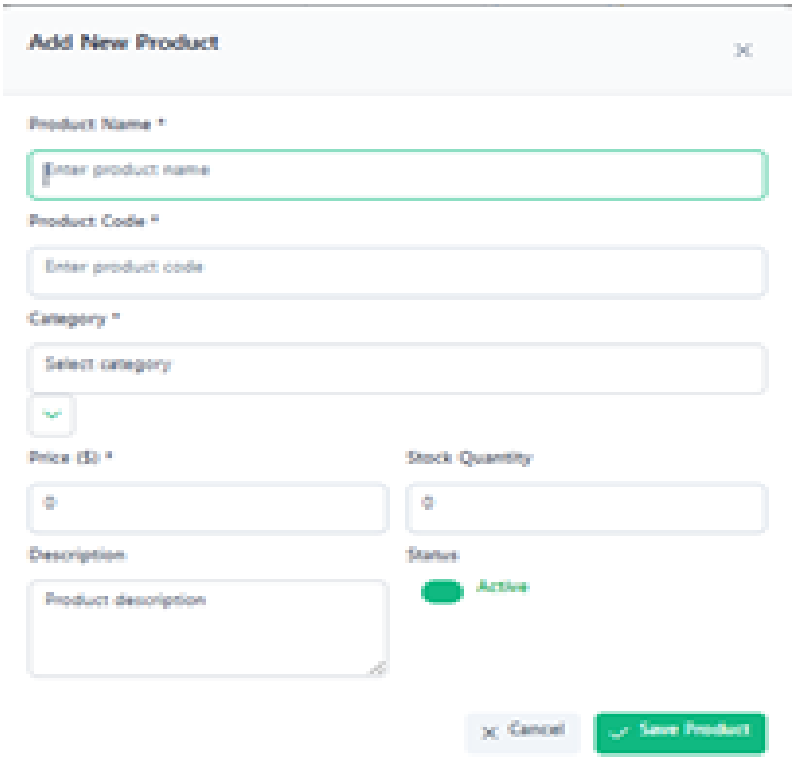


Figure 4.3: Product List Page

4.3.2 Add Product

To create a new product, click Add Product and enter the Product Code, Product Name, and select the appropriate Category from the dropdown. Fill in the Price, set the initial Stock Quantity (if required), and toggle the Active status. Review all the entered information for accuracy, then click Save to submit the record.


Product Management
Manage your products and categories
Debug Structure
Test APIs
Create Sample Data
Manage Categories
Add Product
Products
All products in inventory
ID
Product Name
Code
Category
Price
Stock
Status
Actions
Active
Total 9 products


Add New Product
Product Name *
Enter product name
Product Code *
Enter product code
Category *
Select category
Price ($) *
0
Stock Quantity
0
Description
Product description
Status
Active
Cancel
Save Product

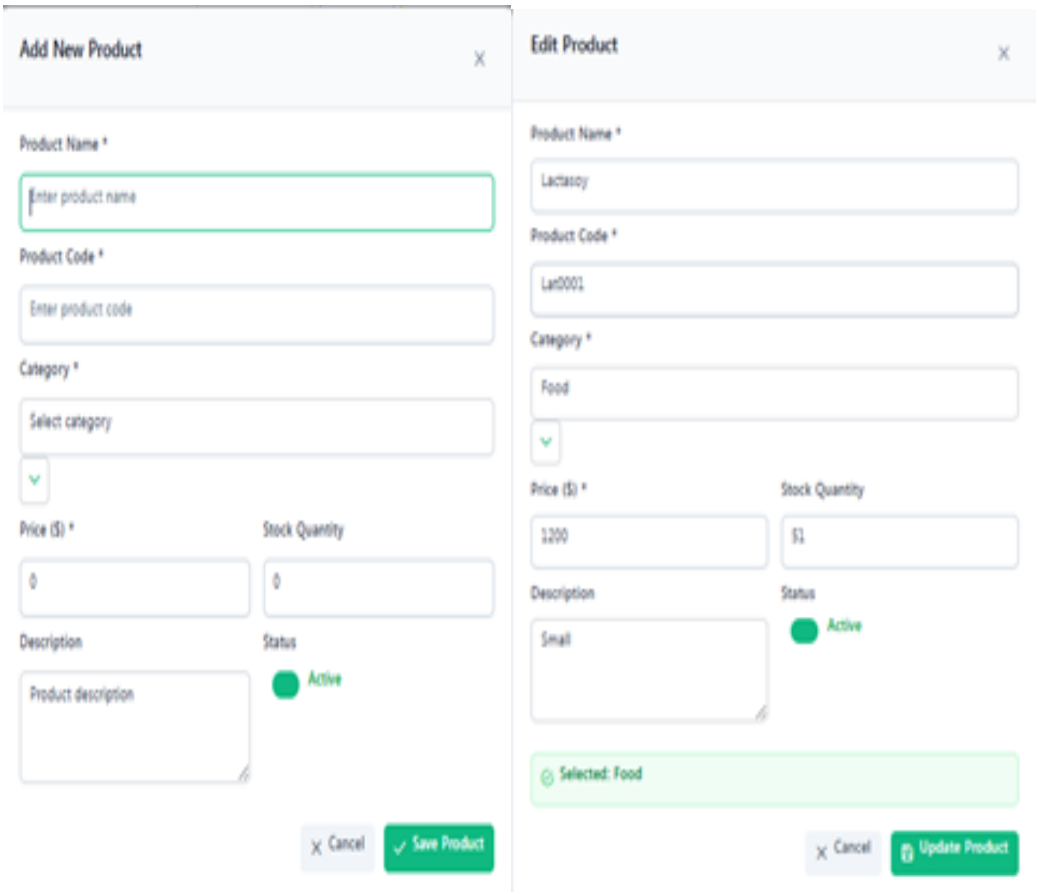
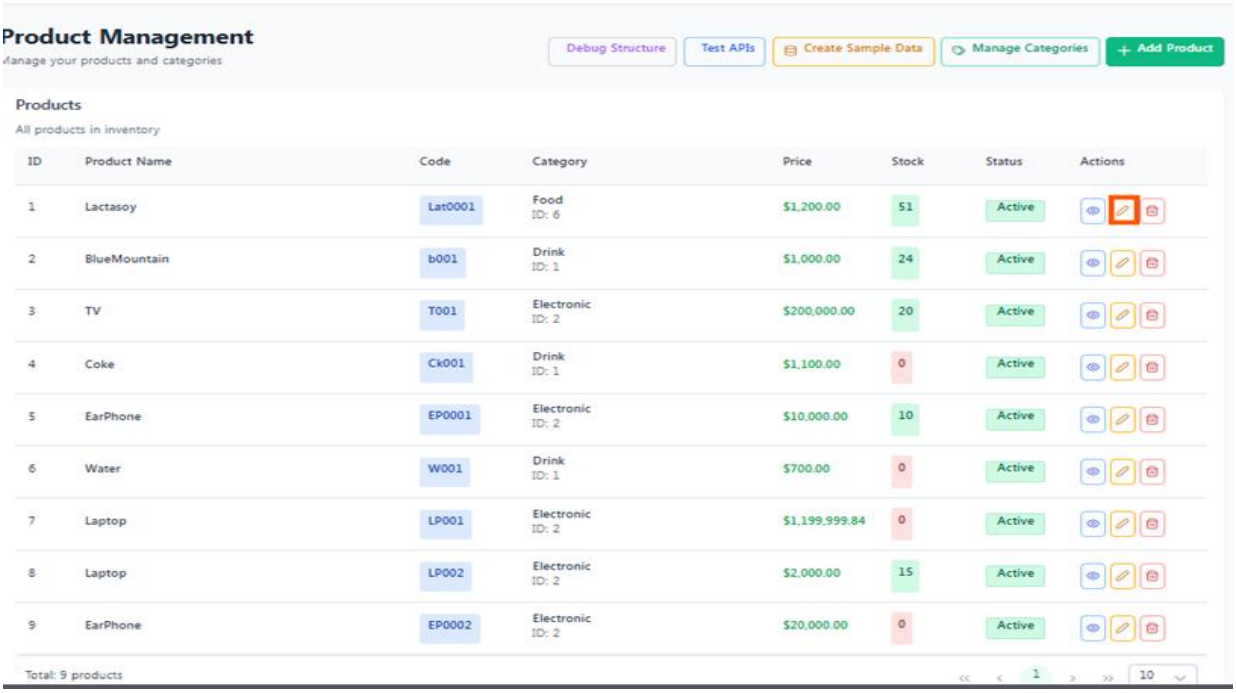


Figure 4.4: Add / Edit Product Form

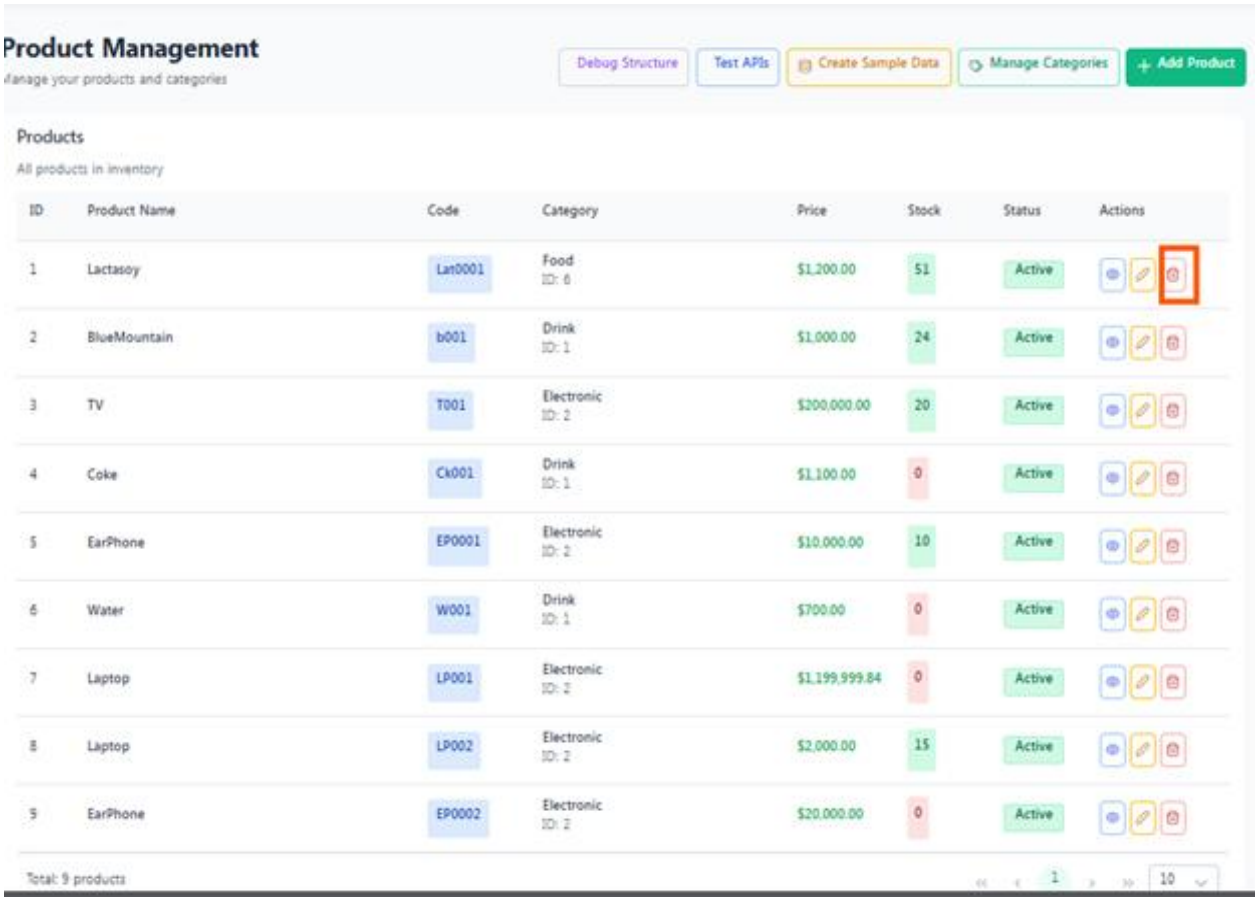
4.3.3 Edit Product

Select the required product from the list and click **Edit**. Update the necessary fields in the form, then click **Save** to apply the changes and verify that the product information has been successfully updated.



4.3.4 Delete Product

Select the product you wish to remove from the list and click **Delete**. Confirm the action when prompted, then verify that the product has been removed from the inventory list or deactivated according to system rules.



4.4 Warehouse Management

The Warehouse Management module maintains the locations where inventory is stored. Warehouse information provides the basis for identifying the storage location of stock.

4.4.1 View Warehouse List

Open the Warehouse Management section to review the list of available warehouses. Check the displayed details for each facility, including the warehouse name and its specific location.

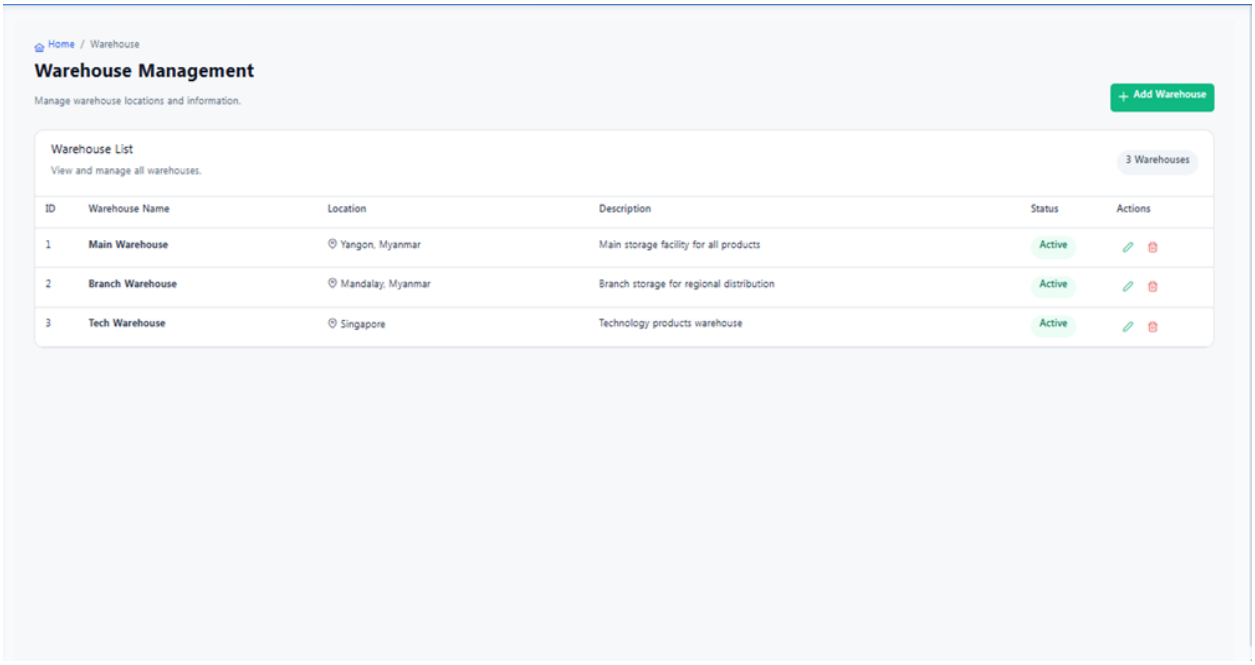
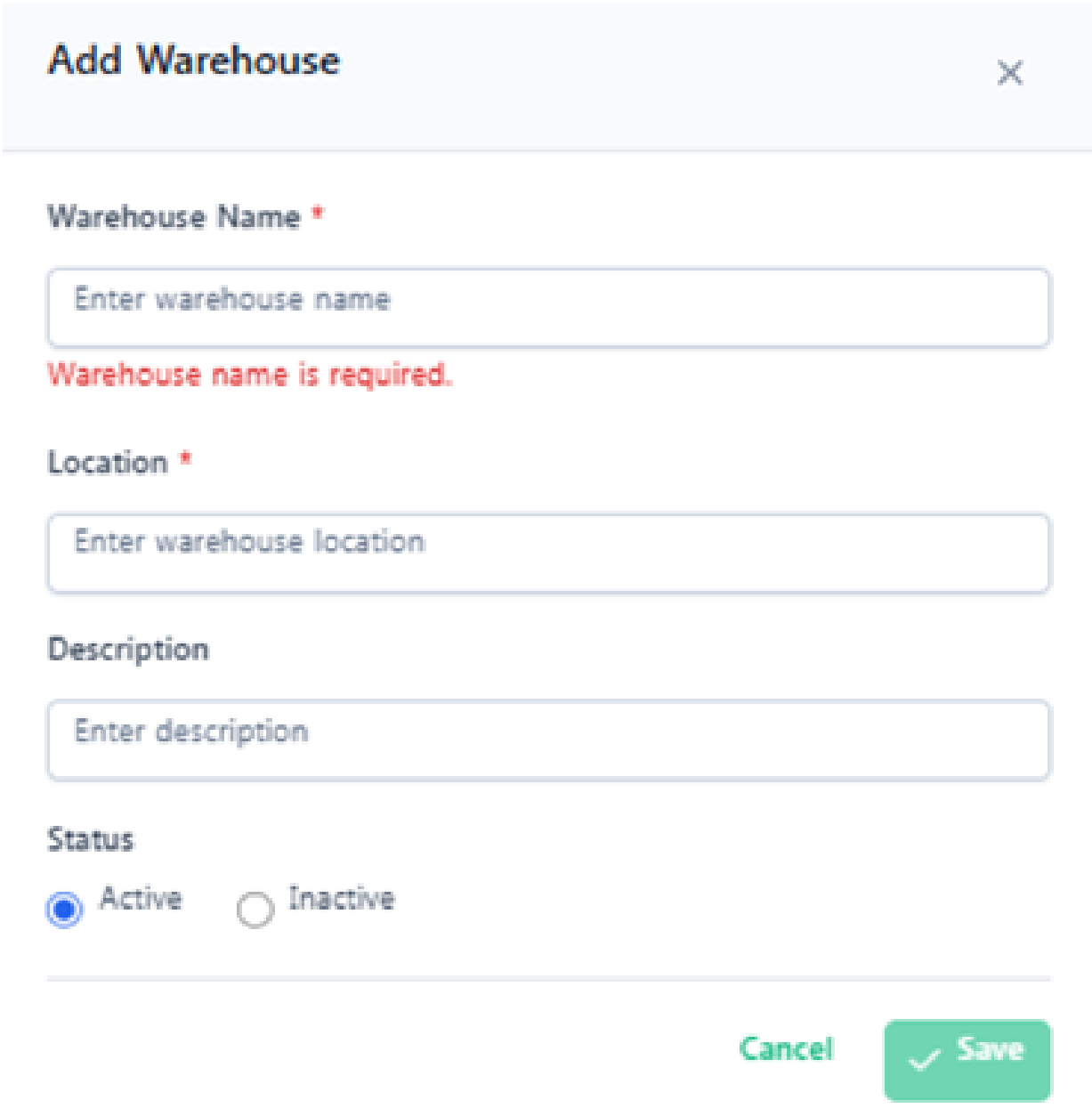


Figure 4.5: Warehouse Management Page

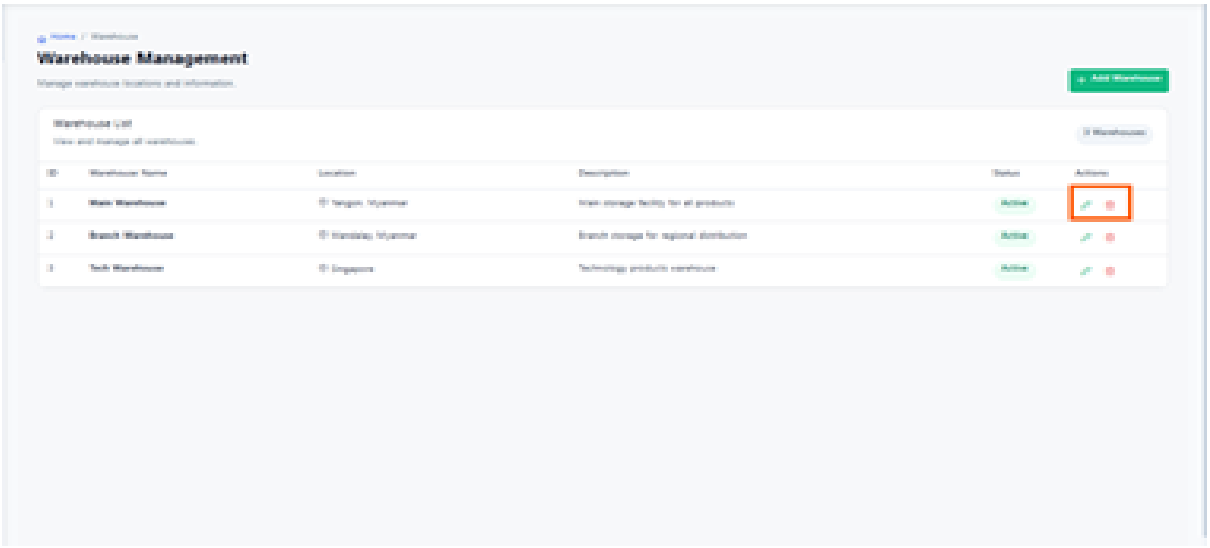
4.4.2 Add Warehouse

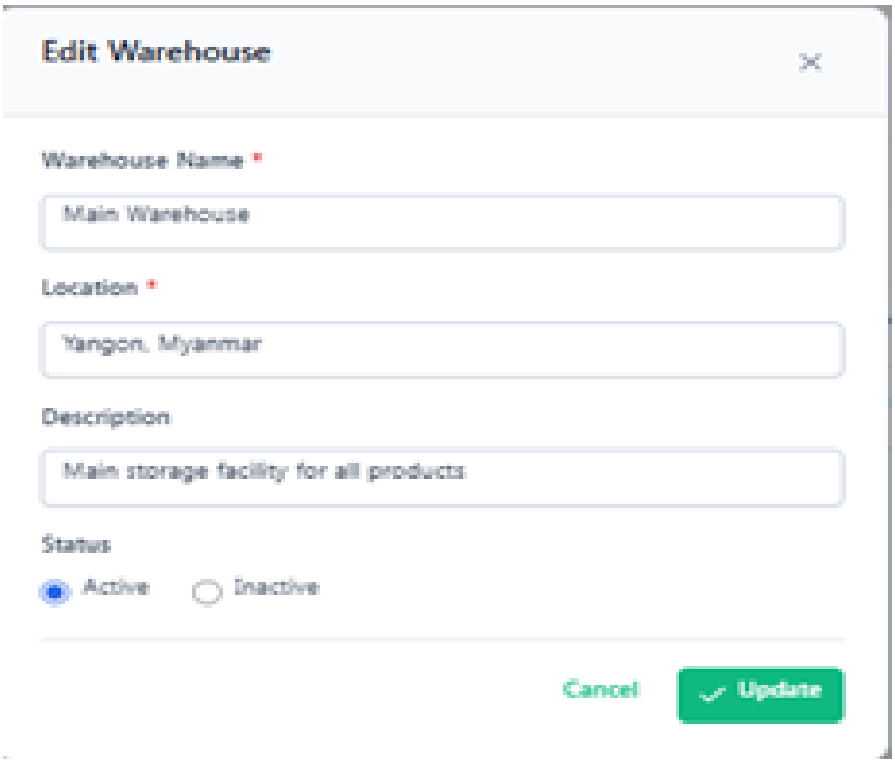


To add a new facility, click Add Warehouse and enter the Warehouse Name along with its Location. Review the entered information for accuracy, then click Save. Once saved, verify that the new warehouse appears in the updated list.

4.4.3 Edit and Delete Warehouse

Select the required warehouse from the list to manage its details. Click **Edit** to update its information, then save the changes. If the warehouse is no longer needed, click **Delete**, confirm the action when prompted, and verify the final warehouse list to ensure the record has been properly updated or removed





4.5 Supplier Management

The Supplier Management module maintains supplier information used during purchasing and incoming stock transactions.

4.5.1 View Supplier List

Open the Supplier Management section to review the complete list of supplier records. You can use the search or filtering options, when available, to quickly find specific suppliers by name, code, or contact details.

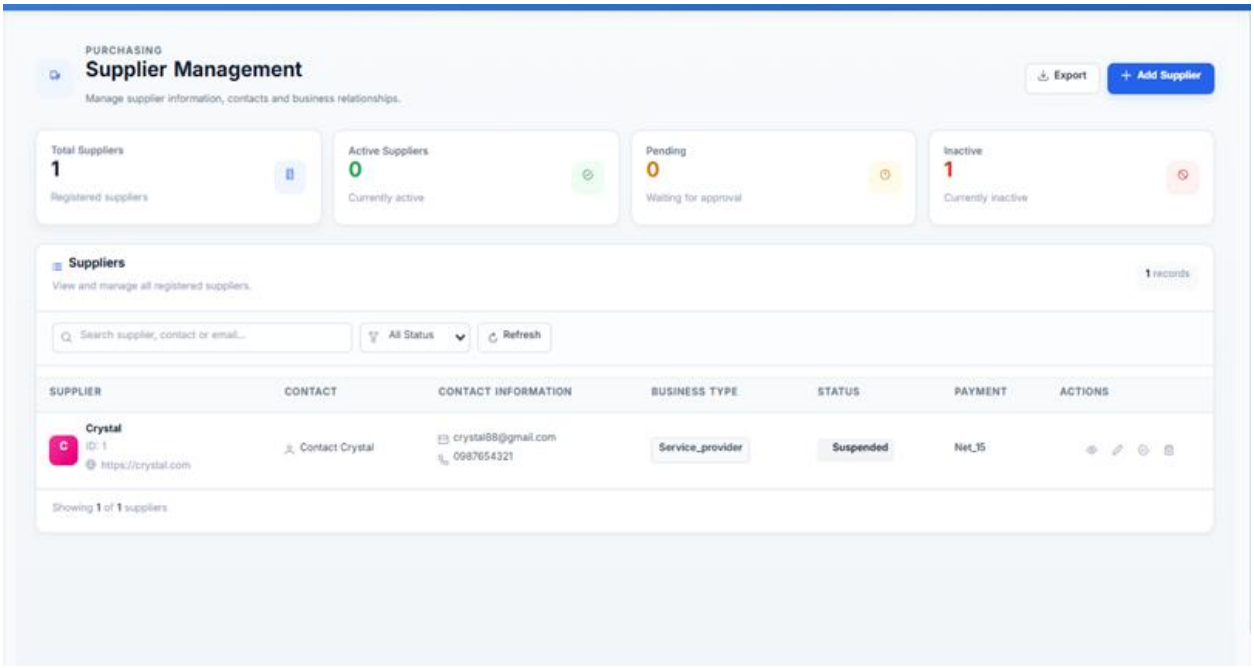
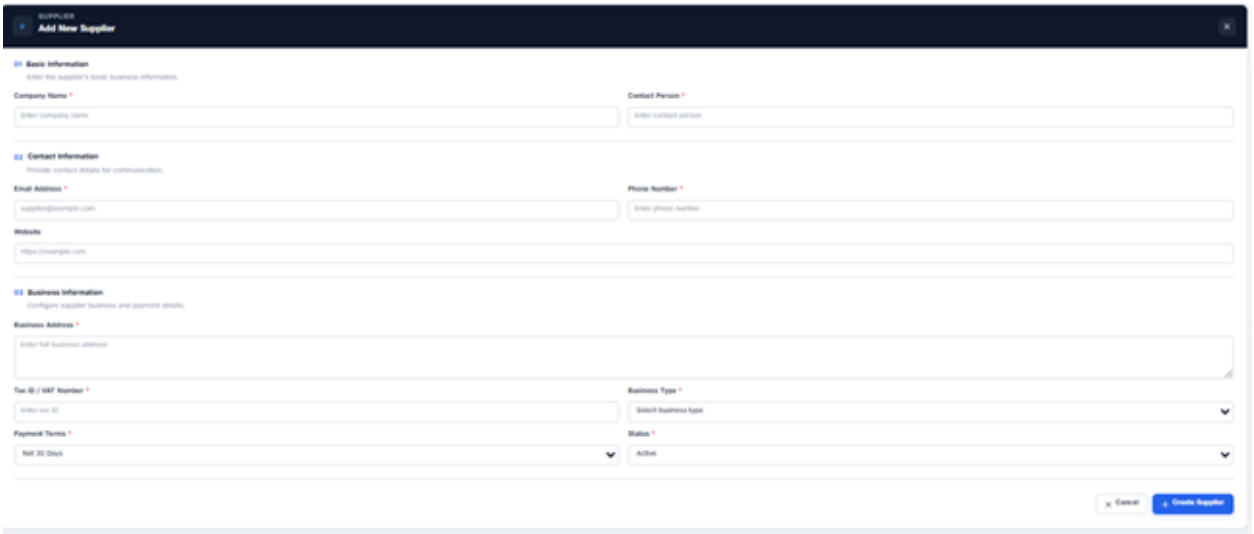


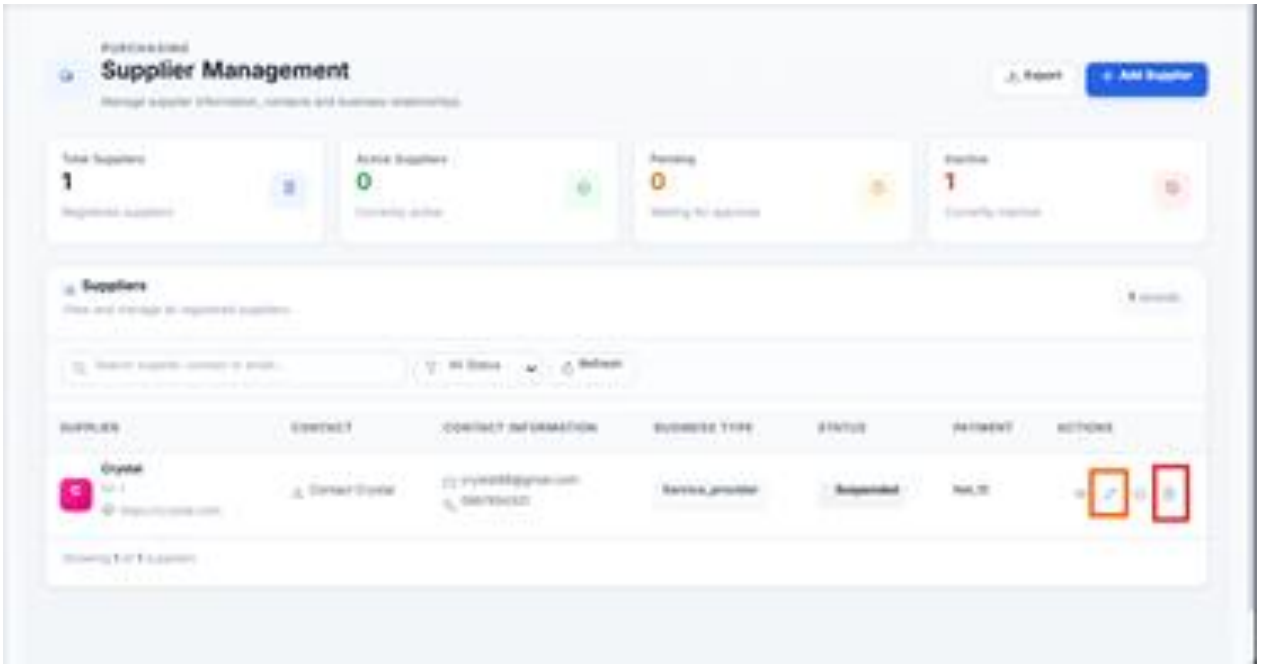
Figure 4.6: Supplier Management Page

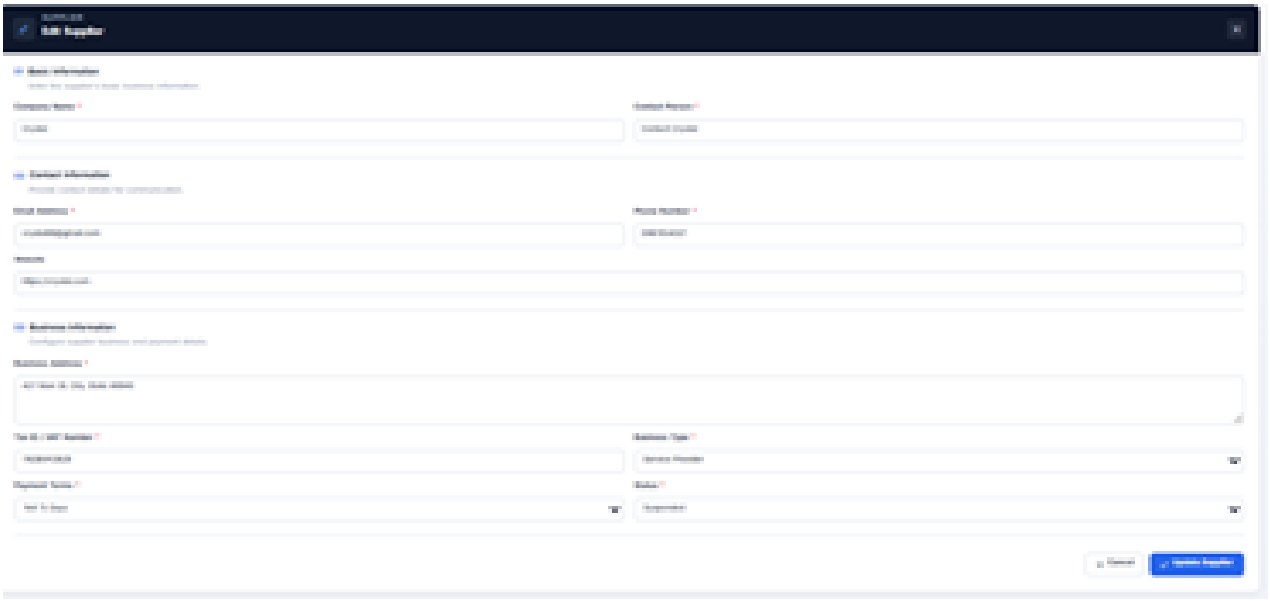
4.5.2 Add Supplier



To register a new supplier, click Add Supplier and enter the Supplier Name, along with the Phone Number and Address when required. Review the information for accuracy, click Save, and verify that the new supplier appears in the list.

4.5.3 Edit and Delete Supplier





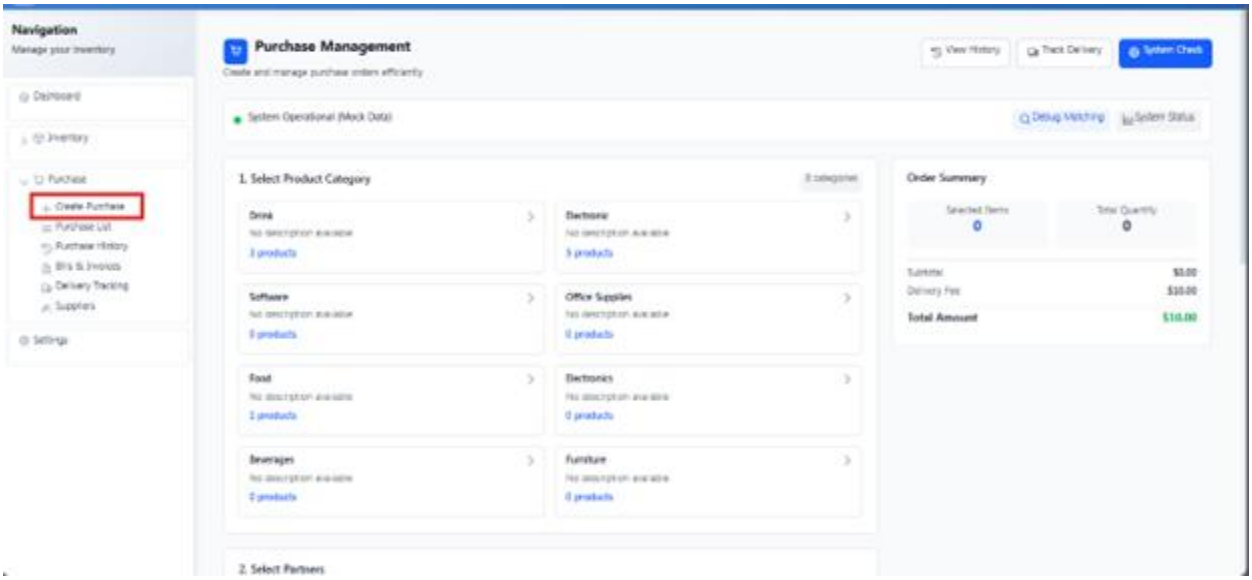
Select the desired supplier from the list to update or remove their details. Click Edit to modify the necessary information and save the changes. If you need to remove the record, click Delete, confirm the action when prompted, and verify that the supplier list reflects the updated records.

4.6 Purchase Management

The Purchase Management module records purchases from suppliers. It connects supplier information with purchased products, quantities, prices, total cost, status, and receipt confirmation.

4.6.1 Create Purchase

Open the Purchase Management section and select the appropriate Supplier. Choose the required product or products, then enter the corresponding Quantity and UnitPrice. Review the automatically calculated Total Cost (or Total Amount) alongside the full purchase details, then submit or click Save to finalize the purchase record.



3. Delivery Information

Hide Delivery

Delivery Method

- **Standard Delivery** — 3-5 business days
- Express Delivery — 1-2 business days
- Same Day — Today
- Pickup — Self-collect

Shipping Company

FedEx

Delivery Address

Enter complete delivery address

Delivery Notes

Special instructions for delivery...

Delivery Method	Estimated Delivery
Standard Delivery	**Sunday, Sep 6**
Shipping Company	Delivery Fee
FedEx	$10.00

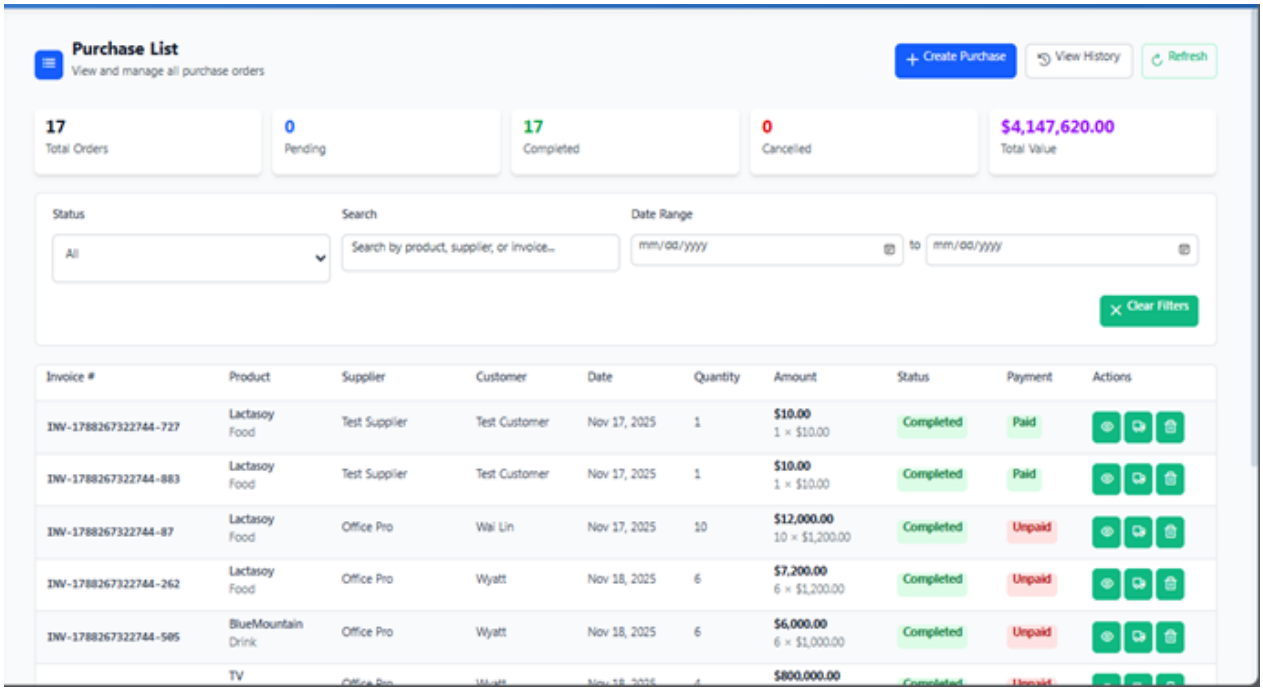
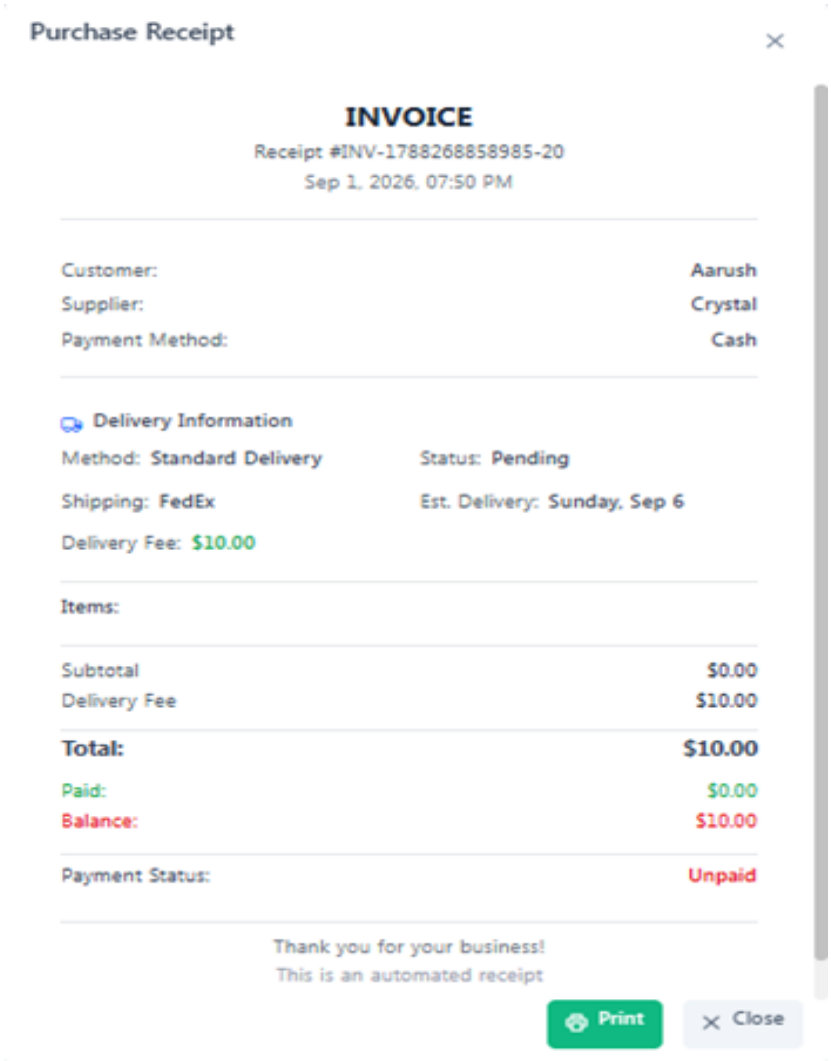


Figure 4.7: Purchase Creation Page / Form

4.6.2 Confirm Purchase Receipt

Open the relevant purchase record to review the supplier, product, quantity, and total amount. Once you have verified that the ordered goods have arrived, complete the receipt confirmation action in the system. Finally, verify that the purchase has automatically increased the related inventory quantity and check the updated stock levels in the system.



If receipt is not confirmed, the purchase should remain in its applicable pending or unreceived status and should not increase available inventory.

4.6.3 Purchase History

Open the Purchase History section to review all previous purchase records. You can check key details such as the supplier, product or category, total amount, order status, and transaction date, as well as use search, filtering, or action controls to locate and manage specific entries efficiently.

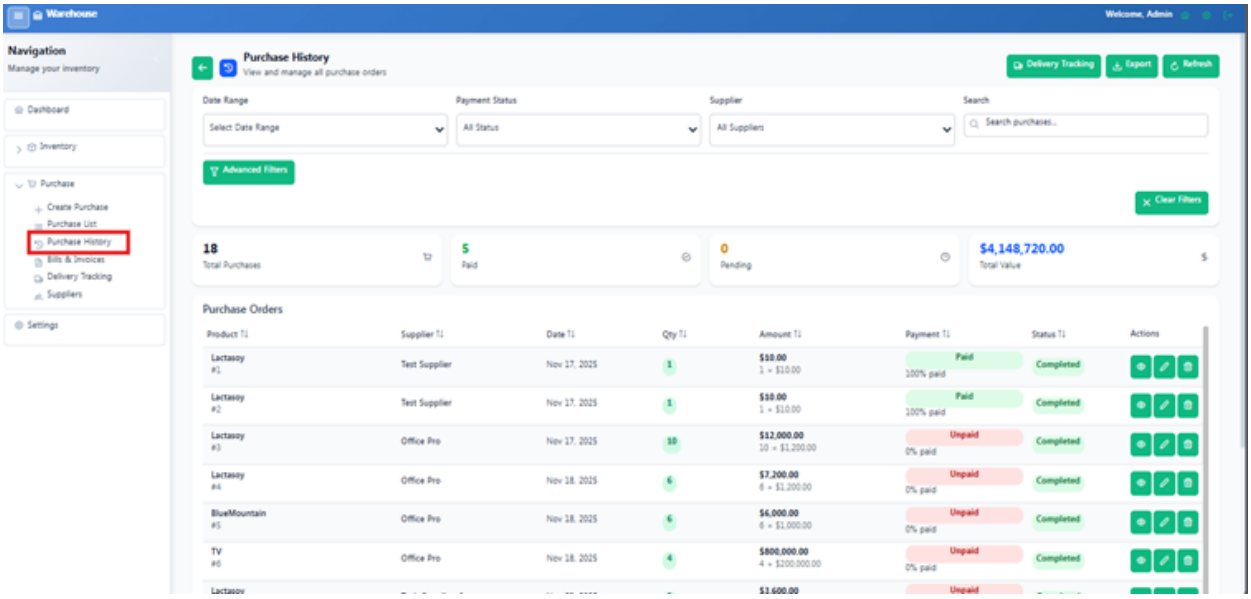


Figure 4.8: Purchase History Page

4.7 Stock Management

Stock Management records the physical movement of products into and out of the warehouse. The main transaction types are Stock IN and Stock OUT.

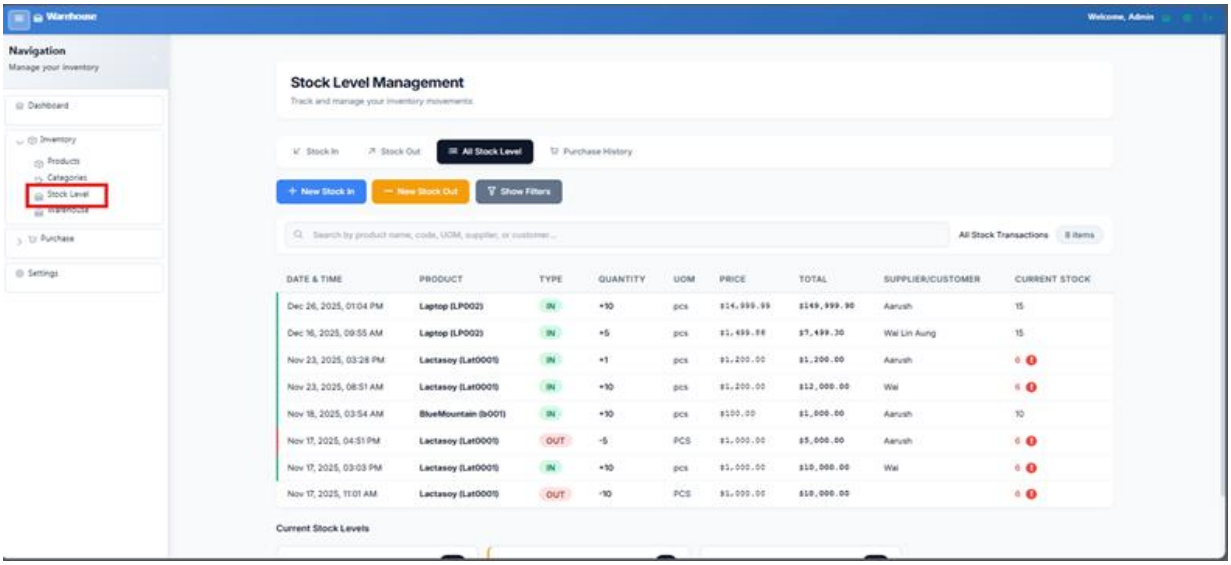


Figure 4.9: Stock Level Management Dashboard

4.7.1 Stock IN

Stock IN is used when products enter the warehouse. A successful Stock IN transaction increases the available stock quantity.

Open the Stock Management section and select **Stock IN**. Choose the target product, then enter the incoming **Quantity** along with the **Supplier**, unit of measure, and price where applicable. Review all transaction details, click **Save** (or **Submit**), and confirm the success notification. Finally, check the product details to verify that the overall stock quantity has been accurately updated.

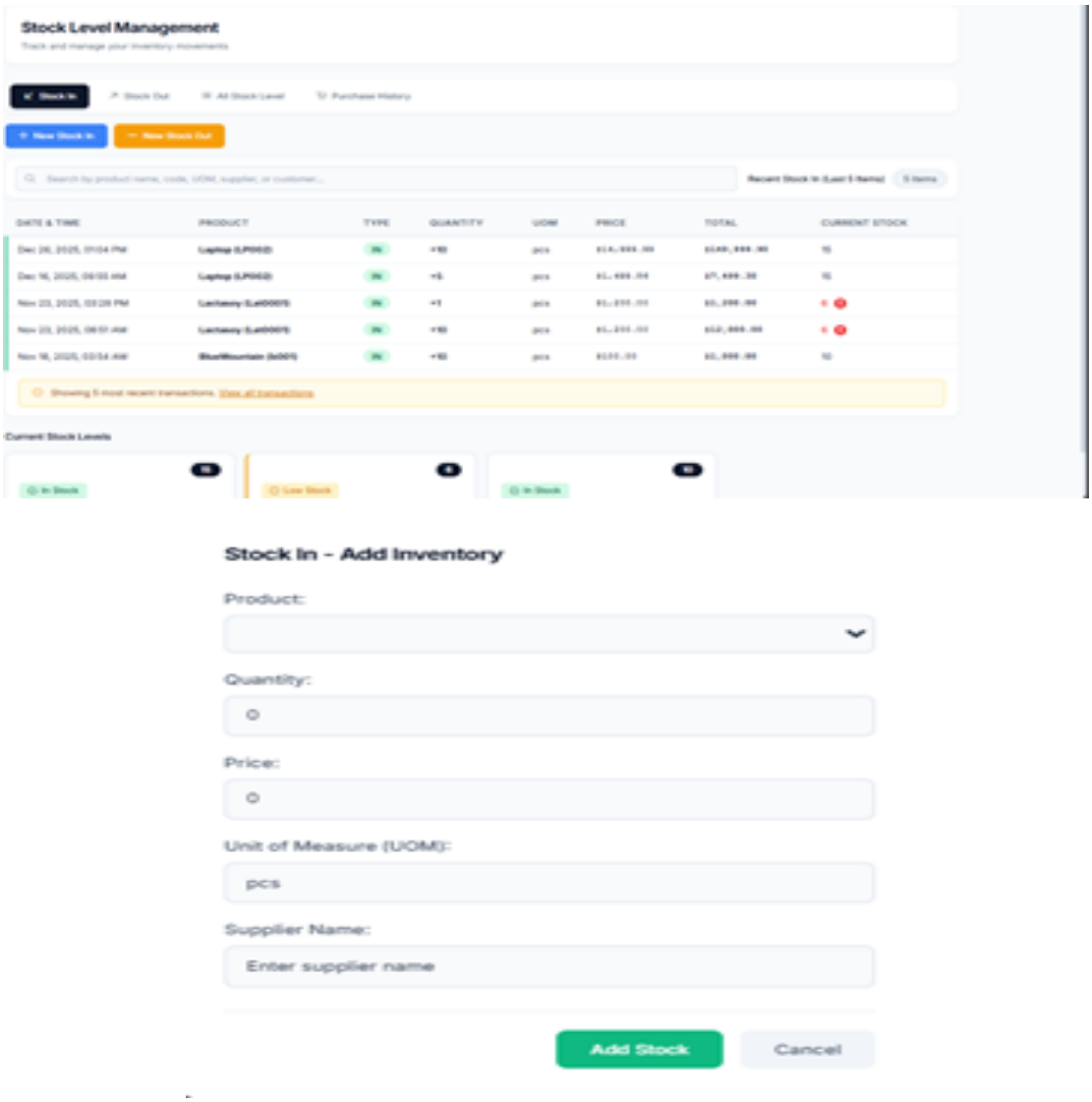
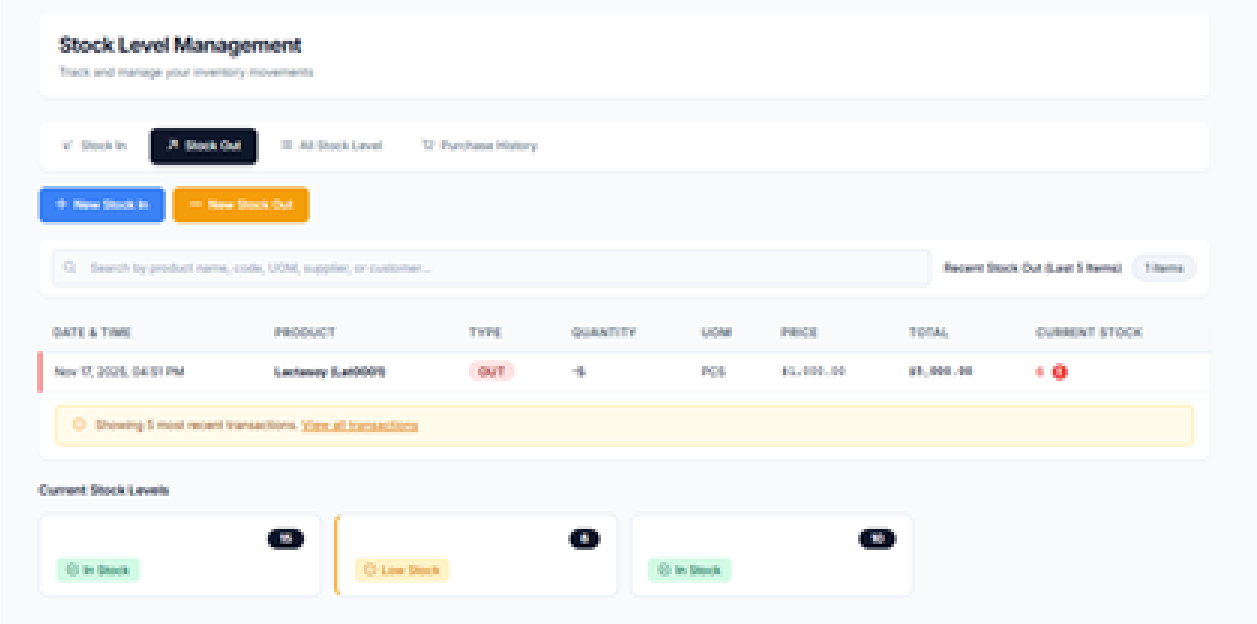


Figure 4.10: New Stock IN Form

4.7.2 Stock OUT

Stock OUT is used when products leave the warehouse. The system checks stock availability before saving the transaction.

Open the Stock Management section and select **Stock OUT**. Choose the target product and enter the **Quantity** to be issued, allowing the system to verify that sufficient stock is available. If stock levels are adequate, enter any required customer information, review the transaction details, and click **Save** (or **Submit**). Finally, verify the success notification and check that the stock quantity has been deducted accordingly.



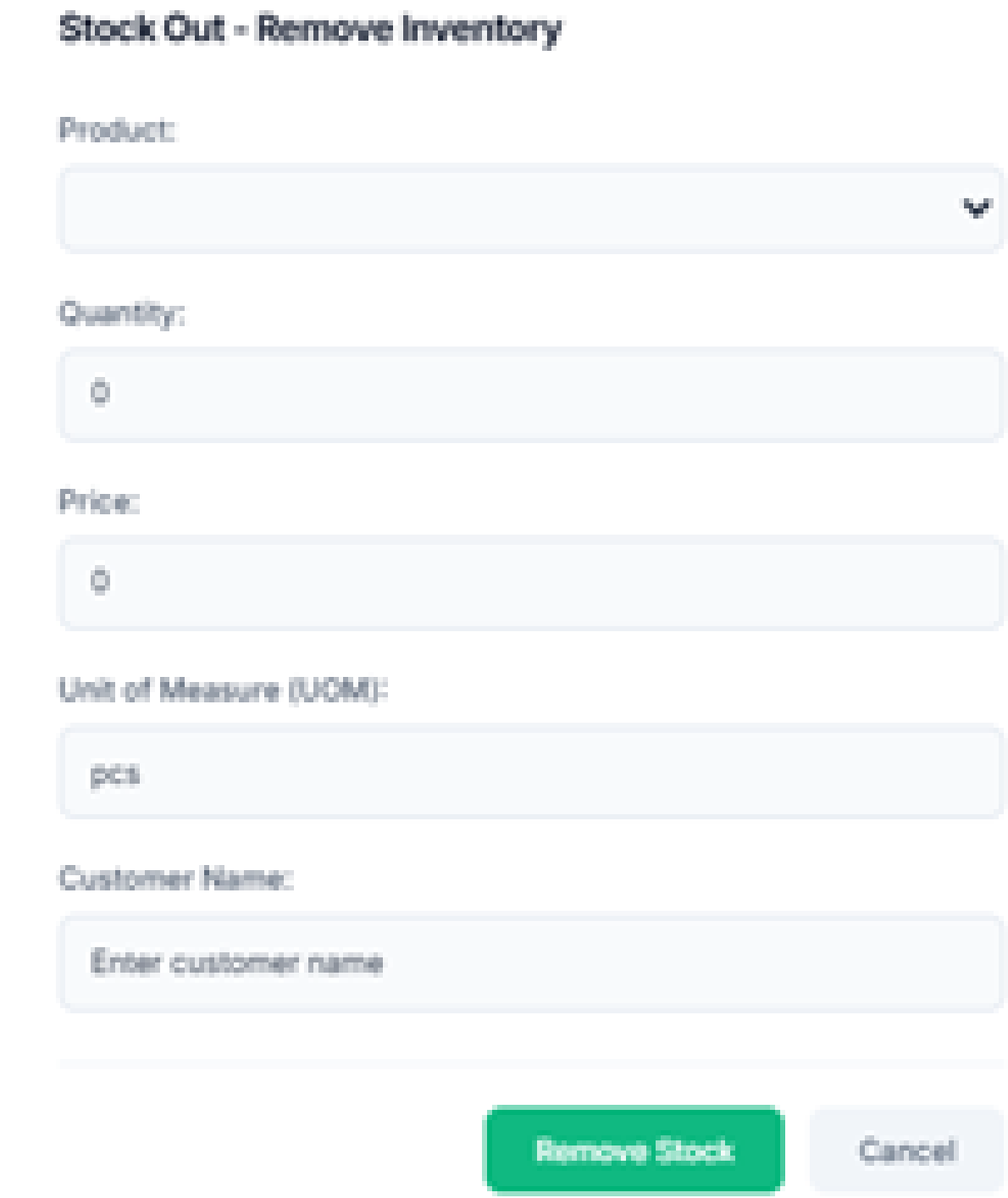


Figure 4.11: New Stock OUT Form

4.7.3 Insufficient Stock Handling

When entering a **Stock OUT** quantity that exceeds available inventory, allow the system to perform its automatic stock check. The system will display an insufficient-stock error notification and prevent submission of the invalid transaction. To proceed, adjust the input to a valid quantity or cancel the process altogether. This safety validation ensures that outgoing transactions never reduce inventory levels below zero.

4.8 Stock Level Monitoring

The Stock Level view provides a current overview of product quantities and helps users identify products requiring replenishment.

Open the Stock Level (or All Stock Level) view to inspect current inventory quantities across all registered products. Review any low-stock indicators highlighted by the system and utilize search or filtering tools to quickly pinpoint specific items. Use these real-time inventory insights to identify depleted stock and determine whether new purchases or Stock IN transactions need to be initiated.

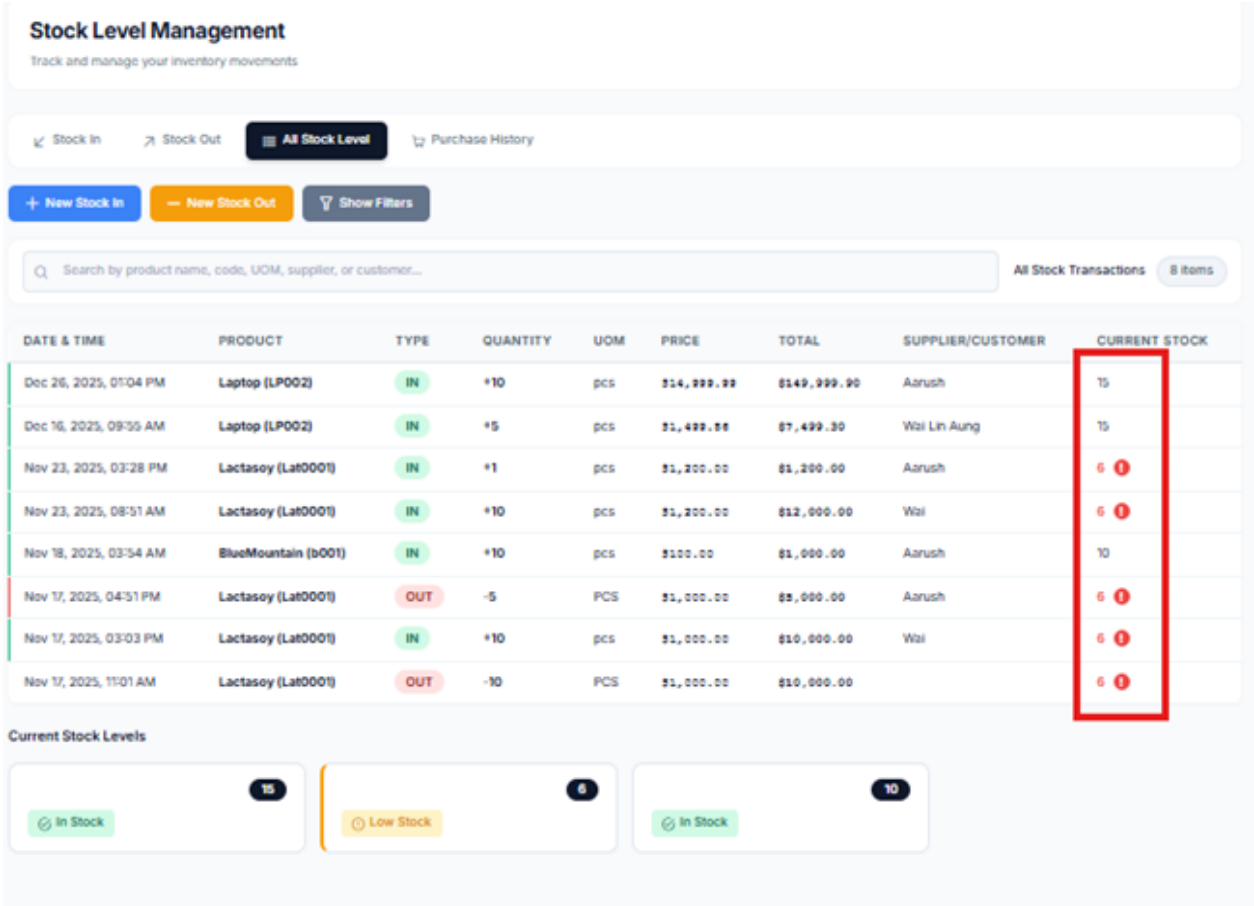


Figure 4.12: Stock Level / All Stock Level View

4.9 Transaction History

Transaction History provides an audit trail of inventory movements and allows users to review previous Stock IN and Stock OUT activities.

Open the Transaction History section to inspect the complete log of inventory movements. Check key details for each entry, including the product name, transaction type (such as Stock IN or Stock OUT), quantity, price, supplier or customer details, and transaction date. Utilize the available search and filtering options to locate specific records and verify historical stock movements across the system.

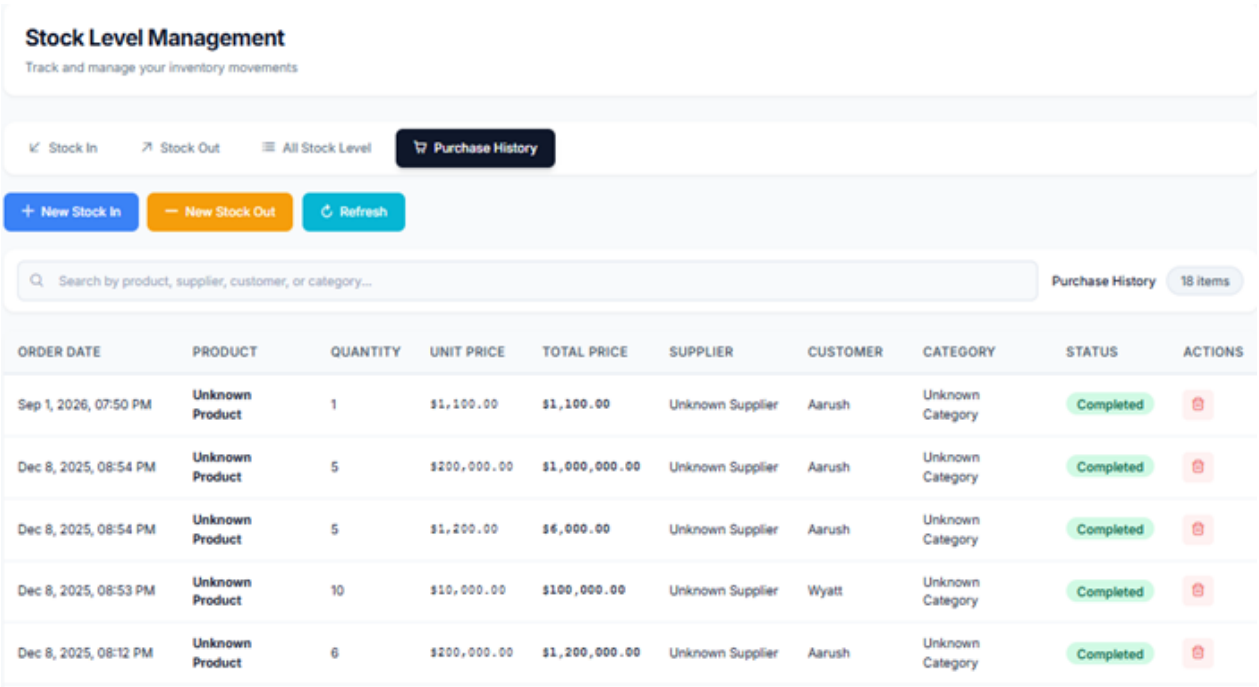


Figure 4.13: Transaction History Page

4.10 End-to-End Warehouse Inventory Workflow

A typical operational workflow begins with master-data setup and continues through purchasing, receiving, stock movement, monitoring, and historical review.

To execute a complete inventory cycle, begin by creating or verifying the required Category, Product (with its assigned category), Warehouse, and Supplier records. Initiate incoming stock by creating a Purchase record, confirming receipt upon arrival, and verifying that total inventory quantities increase accordingly. For subsequent inventory movements, process a Stock IN transaction for additional inbound goods or a Stock OUT transaction when issuing products to customers. Before completing any outgoing issuance, ensure the system validates stock sufficiency to prevent negative balances. Finally, regularly inspect the Stock Level view to monitor current quantities and consult both Transaction History and Purchase History to maintain complete operational traceability.

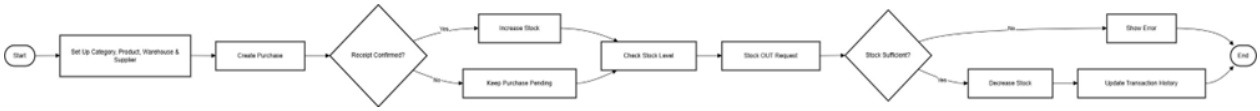


Figure 4.14: End-to-End Warehouse Inventory Workflow

4.11 Success and Error Notifications

The system provides immediate feedback after user actions. Success notifications confirm completed operations, while error notifications explain why an operation cannot be completed.

Table 4.1: System Response and User Actions for Notifications

Situation	Expected System Response	User Action
Category/Product saved	Success message and refreshed list	Verify the record
Purchase saved	Success message and purchase record retained	Review purchase status
Stock IN saved	Success message and increased stock	Verify quantity

Situation	Expected System Response	User Action
Stock OUT with sufficient stock	Success message and decreased stock	Verify quantity
Stock OUT with insufficient stock	Error message and rejected transaction	Enter valid quantity or cancel
Invalid or incomplete form	Validation message	Correct required fields

4.12 Implementation Summary

The implemented modules work together as a single warehouse inventory workflow. Category and Product Management establish the master data used by transactions. Supplier and Warehouse Management provide supporting information for procurement and storage. Purchase Management records procurement activities and can update inventory when receipt is confirmed. Stock IN and Stock OUT maintain inventory movement, while Stock Level and Transaction History provide operational visibility and traceability.

CHAPTER 5

CONCLUSION

5.1 Conclusion

The Inventory Management System provides a structured digital solution for warehouse inventory and purchasing operations. It centralizes product, category, supplier, warehouse, purchase, and stock transaction records and reduces the dependence on manual record keeping. The Stock IN and Stock OUT functions provide a systematic method for maintaining inventory quantities, while stock validation helps prevent invalid outgoing transactions.

The purchase workflow connects procurement activities with inventory management by recording supplier and purchase information and updating stock when a purchase is confirmed as received. Transaction history also improves traceability by allowing users to review previous inventory activities.

Overall, the system establishes a practical foundation for managing warehouse inventory and purchasing data through a web-based architecture using Angular, Spring Boot REST APIs, and MySQL.

5.2 Future Enhancements

- Role-based access control for different types of inventory users.
- Barcode or QR-code support for faster product identification.
- Automated reorder notifications based on configurable minimum stock levels.
- Advanced inventory and purchasing reports with export functions.
- Multi-warehouse stock transfer and consolidated inventory reporting.
- Integration with accounting or supplier systems.